

Emergent Mathematics

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Part I

Harmonic Analysis

1 Hilbert Space

Exercise 1.1 (Finite-dimensional vector spaces and Hilbert spaces). Work over $\mathbb{C}$. Define the tensor product $V \otimes W$ by its universal bilinear map and construct its associativity, unit, and symmetry isomorphisms from that property. A vector space V is called finite-dimensional when there are a vector space $V^\vee$ and maps

$$\text{coev}_V : \mathbb{C} \longrightarrow V \otimes V^\vee, \quad \text{ev}_V : V^\vee \otimes V \longrightarrow \mathbb{C}$$

whose two composites obtained by inserting coev_V and then applying ev_V are the identities of V and $V^\vee$. Let $\sigma_{V, V^\vee}(v \otimes \phi) = \phi \otimes v$ be the symmetry. Define

$$\text{Tr}(T) = \text{ev}_V \circ \sigma_{V, V^\vee} \circ (T \otimes 1) \circ \text{coev}_V$$

for $T \in \text{End}(V)$. Prove cyclicity of trace and the direct-sum and tensor product formulas. Define a complex inner-product space, its norm, Cauchy sequences, and completion. Define a Hilbert space to be a complete inner-product space and prove the projection theorem for a closed subspace. Deduce the Riesz representation theorem for bounded linear functionals. For a bounded operator T define its adjoint intrinsically by $\langle Tx, y \rangle = \langle x, T^*y \rangle$, and prove $\|T^*T\| = \|T\|^2$.

Exercise 1.2 (C^* -algebras, states, and representations). Define a unital C^* -algebra to be a complete normed complex algebra A with an involution satisfying $\|a^*a\| = \|a\|^2$. Define positive elements, states, and $*$ -representations. Prove that every positive functional ω satisfies Cauchy–Schwarz. On A/N_ω , where $N_\omega = \{a : \omega(a^*a) = 0\}$, complete

$$\langle [a], [b] \rangle_\omega = \omega(b^*a)$$

and show that left multiplication gives a cyclic representation $(\pi_\omega, \mathcal{H}_\omega, \Omega_\omega)$ with $\omega(a) = \langle \pi_\omega(a)\Omega_\omega, \Omega_\omega \rangle$. Show that the direct sum over all states is faithful.

Exercise 1.3 (Gelfand transform and the spectral theorem). For a unital commutative C^* -algebra A , define its character space $\hat{A}$ and the Gelfand transform $a \mapsto \hat{a}$. Prove, by maximal ideals and the C^* -identity, that $A \simeq C(\hat{A})$. Prove the Riesz–Markov theorem for positive functionals on $C(X)$ by constructing a regular Borel measure through outer approximation. For a bounded self-adjoint operator H , apply this to $C^*(1, H)$ and construct the unique projection-valued measure P on $\text{spec}(H)$ satisfying

$$f(H) = \int_{\text{spec}(H)} f(E) \, dP(E), \quad H = \int_{\text{spec}(H)} E \, dP(E).$$

Prove the bounded Borel functional calculus, spectral mapping, and $\langle f(H)x, y \rangle = \int f \, d\mu_{x,y}$, where $\mu_{x,y}(B) = \langle P(B)x, y \rangle$.

Exercise 1.4 (Closed operators and the unbounded spectral theorem). For a densely defined operator D on a Hilbert space, define its graph, adjoint, closure, resolvent, and the conditions of being symmetric, self-adjoint, and essentially self-adjoint. Prove the closed-graph criterion for the

adjoint and show that $(D - i)(D + i)^{-1}$ is unitary exactly when D is self-adjoint. Apply the bounded spectral theorem to this Cayley transform and construct the unique projection-valued measure satisfying

$$D = \int_{\mathbb{R}} \lambda dP(\lambda), \quad \text{Dom}(D) = \left\{ x : \int_{\mathbb{R}} \lambda^2 d\mu_x(\lambda) < \infty \right\}.$$

Develop the measurable functional calculus, including its domain rule and spectral mapping. Define a densely defined closed semibounded quadratic form and prove the representation theorem that associates it with a self-adjoint operator. For a semibounded symmetric operator, close its form and construct the Friedrichs extension. Prove that compact resolvent is equivalent to a discrete spectral decomposition with finite-rank spectral projections and no finite accumulation point.

Exercise 1.5 (Holomorphic functions and contour integrals). Define complex differentiability, holomorphic and meromorphic functions, piecewise smooth contour integrals, winding number, and residues. Prove Goursat's theorem on triangles, the Cauchy integral formula, local power-series expansion, the identity and maximum principles, and the residue theorem for a finite collection of polygonal boundary curves. Derive the argument principle and Schwarz reflection. Define analytic continuation along a path by overlapping power-series germs, prove uniqueness, and identify monodromy on a simply connected domain. For a holomorphic family of bounded invertible operators, prove that inversion is holomorphic and that a pole of a finite-rank resolvent has residue equal to its spectral projection.

Exercise 1.6 (Green functions and spectral measures). For $z \notin \text{spec}(H)$ define the resolvent $R_H(z) = (H - z)^{-1}$ and the scalar Green function

$$G_x(z) = \langle R_H(z)x, x \rangle = \int \frac{d\mu_x(E)}{E - z}.$$

Prove the resolvent identity, $\Im G_x(z) > 0$ for $\Im z > 0$, and the Stieltjes inversion formula

$$\mu_x((a, b)) = \lim_{\epsilon \downarrow 0} \frac{1}{\pi} \int_a^b \Im G_x(E + i\epsilon) dE$$

when the endpoints are not atoms. Conversely, prove the Herglotz representation for a function with positive imaginary part and the decay $G(iy) = O(y^{-1})$. Thus identify scalar Green functions as the Stieltjes special-function transforms of finite positive measures.

Exercise 1.7 (Green-function resonance spectroscopy). Let H act on a finite-dimensional Hilbert space, let d be a normalized state prepared by a weak monochromatic probe, and let $\gamma > 0$ be its energy resolution. Define the absorption profile

$$A_\gamma(E) = \frac{1}{\pi} \Im G_d(E + i\gamma).$$

Use the spectral theorem to prove

$$A_\gamma(E) = \sum_{\lambda \in \text{spec}(H)} \|P_{\{\lambda\}}d\|^2 \frac{\gamma/\pi}{(E - \lambda)^2 + \gamma^2}.$$

Show that an isolated line is centered at λ , has full width at half maximum 2γ , integrated intensity $\|P_{\{\lambda\}}d\|^2$, and converges to the corresponding spectral mass as $\gamma \downarrow 0$. Formulate the experimental test as simultaneous comparison of line positions, widths, and oscillator strengths with these three quantities.

2 Fourier Analysis

Exercise 2.1 (Locally compact groups and Haar measure). Let G be a locally compact group. Construct a nonzero positive left-translation-invariant functional on $C_c(G)$ by the covering-number argument, construct its regular Borel measure by outer approximation, and extend the preceding Riesz–Markov representation to $C_c(G)$ and prove the uniqueness needed to obtain left Haar measure dx , unique up to scale. Construct right Haar measure and the continuous homomorphism $\Delta_G : G \rightarrow \mathbb{R}_{>0}$ characterized by

$$\int_G f(xg) dx = \Delta_G(g)^{-1} \int_G f(x) dx.$$

Define convolution and involution by

$$(f * g)(x) = \int_G f(y)g(y^{-1}x) dy, \quad f^*(x) = \Delta_G(x^{-1})\overline{f(x^{-1})},$$

and prove that $L^1(G)$ is a Banach $*$ -algebra and that left translation is unitary on $L^2(G)$. Prove that $\Delta_G = 1$ for abelian and compact groups; in the abelian case adopt additive notation, so that the formulas reduce to the earlier additive convolution and reflection involution.

Exercise 2.2 (Pontryagin duality and Fourier decomposition). Define $\widehat{G} = \text{Hom}_{\text{cont}}(G, U(1))$ with the compact-open topology. Normalize Haar measure on $\widehat{G}$ so that

$$\widehat{f}(\chi) = \int_G f(x)\overline{\chi(x)} dx, \quad f(x) = \int_{\widehat{G}} \widehat{f}(\chi)\chi(x) d\chi$$

for the functions obtained by convolving two compactly supported continuous functions. Extend the transform to a unitary $L^2(G) \rightarrow L^2(\widehat{G})$, prove Plancherel and convolution–multiplication, and prove that evaluation identifies G with $\widehat{\widehat{G}}$. Specialize all normalizations to $\mathbb{R}^n$, $\mathbb{Z}^n$, and $\mathbb{T}^n$.

Exercise 2.3 (Poisson summation and Gaussian Fourier kernels). On $\mathbb{R}^n$ use

$$\widehat{f}(\xi) = \int_{\mathbb{R}^n} f(x)e^{-2\pi i x \cdot \xi} dx.$$

Define the Schwartz space by its seminorms, prove Fourier inversion there, and compute $\widehat{e^{-\pi t|x|^2}} = t^{-n/2}e^{-\pi|\xi|^2/t}$. For a lattice L define its covolume and dual lattice intrinsically and prove

$$\sum_{x \in L} f(x) = \frac{1}{\text{covol}(L)} \sum_{\xi \in L^*} \widehat{f}(\xi).$$

Derive the Jacobi transformation $\vartheta(-1/\tau) = (-i\tau)^{1/2}\vartheta(\tau)$ for $\vartheta(\tau) = \sum_{m \in \mathbb{Z}} e^{\pi i m^2 \tau}$, fixing the square-root branch by positivity on the imaginary axis.

Exercise 2.4 (Dirichlet, Fejér, and sinc Fourier kernels). On $\mathbb{T}$ define

$$D_N(\theta) = \sum_{n=-N}^N e^{in\theta} = \frac{\sin((N+1/2)\theta)}{\sin(\theta/2)}, \quad F_N = \frac{1}{N+1} \sum_{j=0}^N D_j.$$

Prove that $F_N \geq 0$, has integral one, and is an approximate identity. Deduce Fejér convergence for continuous functions and the L^2 convergence of Fourier sums. Define $\text{sinc } u = \sin(u)/u$ continuously at zero and compute it as the normalized Fourier transform of an interval.

Exercise 2.5 (Fraunhofer diffraction by rectangular and circular apertures). For monochromatic scalar light of wavelength λ , derive from the Huygens integral in the far-field limit that the amplitude at angle θ is the Fourier transform of the aperture evaluated at θ/λ . Deduce the rectangular-aperture intensity as a product of squared sinc functions. For N equally spaced identical slits, derive the squared Dirichlet-kernel array factor and its peak spacing. Define $J_1(u) = \sum_{k \geq 0} (-1)^k (u/2)^{2k+1} / (k!(k+1)!)$, derive its differential equation and polar integral, and for a circular aperture of diameter D deduce

$$\frac{I(\theta)}{I(0)} = \left[\frac{2J_1(\pi D \sin \theta / \lambda)}{\pi D \sin \theta / \lambda} \right]^2.$$

Show from the first positive zero $j_{1,1} = 3.831705\dots$ that the first dark ring obeys $\sin \theta = 1.21967\dots \lambda/D$, and state the measured quantity needed to test this prediction.

3 Compact Harmonic Analysis

Exercise 3.1 (Compact groups and unitary representations). Use the preceding Haar construction and compactness to normalize Haar measure on a compact group K to total mass one. Average an inner product to prove that every finite-dimensional continuous representation is unitarizable and every invariant subspace has an invariant orthogonal complement. Define irreducibility, intertwiners, contragredients, and characters. Prove Schur's lemma and character orthogonality without choosing coordinates.

Exercise 3.2 (Peter–Weyl decomposition). For an irreducible unitary representation π define its matrix-coefficient map

$$C_\pi : V_\pi \otimes V_\pi^\vee \longrightarrow C(K), \quad C_\pi(v \otimes \phi)(g) = \phi(\pi(g^{-1})v).$$

Prove Schur orthogonality and that $\sqrt{\dim V_\pi} C_\pi$ is an isometry. Use convolution, finite-rank spectral projections, and Stone–Weierstrass to prove the unitary decomposition

$$L^2(K) \simeq \widehat{\bigoplus_{\pi \in \widehat{K}} V_\pi \otimes V_\pi^\vee},$$

with left translation on V_π and right translation on $V_\pi^\vee$. Prove the required Stone–Weierstrass density statement from separation of points and closure under conjugation. Derive Fourier inversion and Plancherel for K .

Exercise 3.3 ($SU(2)$, Wigner functions, and spherical harmonics). Define $SU(2)$ and prove that conjugation on its traceless Hermitian endomorphisms gives the two-to-one covering $SU(2) \rightarrow SO(3)$. Construct the permutation coinvariant quotient $\text{Sym}^n(V)$ of $V^{\otimes n}$. Construct the irreducibles $V_j = \text{Sym}^{2j}(\mathbb{C}^2)$ for $j \in \frac{1}{2}\mathbb{Z}_{\geq 0}$ and define the Wigner functions $D_{mn}^j(g)$ as their distinguished weight matrix coefficients. Derive their Euler-angle formula, orthogonality, completeness, and

$$\chi_j(\theta) = \frac{\sin((2j+1)\theta/2)}{\sin(\theta/2)} = U_{2j}(\cos(\theta/2)),$$

thereby defining the Chebyshev polynomials U_n . Identify the spherical harmonics with the right- $U(1)$ -invariant coefficients and derive the addition kernel

$$\sum_{m=-\ell}^{\ell} Y_{\ell m}(x) \overline{Y_{\ell m}(y)} = \frac{2\ell+1}{4\pi} P_\ell(x \cdot y).$$

Exercise 3.4 (Clebsch–Gordan and Wigner special functions). Prove $V_{j_1} \otimes V_{j_2} \simeq \bigoplus_{j=|j_1-j_2|}^{j_1+j_2} V_j$. Define Clebsch–Gordan coefficients as the normalized components of these canonical isometric inclusions relative to the distinguished weight lines, and define the Wigner $3j$ and $6j$ symbols from them. Derive their orthogonality, the triangle rules, the Racah recoupling identity, and the integral of three Wigner functions. Prove the Wigner–Eckart theorem by identifying tensor operators with intertwiners.

Exercise 3.5 (Rotational spectroscopy). Let a rigid symmetric-top molecule have moment-of-inertia operator with principal values $I_\perp, I_\perp, I_\parallel$. Quantize its rotational Hamiltonian by the $SU(2)$ action and prove

$$E_{JK} = \frac{\hbar^2}{2I_\perp} J(J+1) + \frac{\hbar^2}{2} (I_\parallel^{-1} - I_\perp^{-1}) K^2.$$

Express electric-dipole transition amplitudes through Wigner $3j$ symbols and derive $\Delta J = 0, \pm 1$, the prohibition $J = 0 \rightarrow J = 0$, and the appropriate ΔK rule for dipoles parallel or perpendicular to the symmetry axis. For a linear rotor deduce the line positions $\nu_{J \rightarrow J+1} = \hbar(J+1)/(2\pi I)$ and the relative line strengths from the $3j$ symbols; these are the quantitative spectroscopic predictions.

Part II

Lie Theory

4 Reductive Groups

Exercise 4.1 (Lie algebras, enveloping algebras, and Hopf algebras). Define a complex Lie algebra, Lie ideals, representations, and the adjoint representation. Construct the tensor algebra by its universal property and define

$$U(\mathfrak{g}) = T(\mathfrak{g}) / \langle x \otimes y - y \otimes x - [x, y] \rangle$$

by its universal property among associative algebras receiving a Lie map from $\mathfrak{g}$. Filter $U(\mathfrak{g})$ by tensor degree, construct the canonical graded map $\text{Sym}(\mathfrak{g}) \rightarrow \text{gr } U(\mathfrak{g})$, and prove it is an isomorphism by the rewriting filtration and the Jacobi identity. Deduce the Poincaré–Birkhoff–Witt theorem, induction and restriction of modules, and the construction of a Verma module from a triangular decomposition. Define coalgebras, bialgebras, and Hopf algebras by their universal diagrams, and construct the cocommutative Hopf structure on $U(\mathfrak{g})$ for which every $x \in \mathfrak{g}$ is primitive.

Exercise 4.2 (Root data and Weyl groups). Define an affine complex algebraic group by its finitely generated reduced commutative coordinate Hopf algebra, after defining nilpotent elements and reduced algebras. Define connectedness, unipotent groups, the unipotent radical, and reductivity, and construct its Lie algebra from derivations at the counit. For a connected complex reductive group G , define a maximal torus T , its character and cocharacter lattices, roots, coroots, and the Weyl group $W = N_G(T)/T$. Derive the root-space decomposition of $\mathfrak{g}$ and the rank-one $\mathfrak{sl}_2$ subalgebra associated with each root. Choose a positive system, define simple roots, dominant weights, the Weyl length, and ρ , and prove the chamber and Bruhat decompositions.

Exercise 4.3 (Highest-weight decomposition). Define weight spaces without choosing vectors in them. Prove that every finite-dimensional irreducible G -representation has a unique dominant

highest weight. For a Borel algebra $\mathfrak{b} = \mathfrak{t} \oplus \mathfrak{n}_+$ define $M_\lambda = U(\mathfrak{g}) \otimes_{U(\mathfrak{b})} \mathbb{C}_\lambda$ and construct the irreducible representation as its unique irreducible quotient. Use the rank-one strings to prove finite dimensionality exactly for dominant integral weights, complete reducibility, and the highest-weight classification.

Exercise 4.4 (Weyl characters and Schur functions). Define exterior powers as alternating tensor quotients and define the determinant of an endomorphism as the scalar by which it acts on the highest nonzero exterior power. Define a partition as a finite weakly decreasing sequence of nonnegative integers. In the group ring of the weight lattice define $A_\mu = \sum_{w \in W} (-1)^{\ell(w)} e^{w\mu}$. Construct the BGG resolution from Verma modules and use its Euler character to prove

$$\text{ch } V_\lambda = \frac{A_{\lambda+\rho}}{A_\rho}, \quad A_\rho = e^\rho \prod_{\alpha > 0} (1 - e^{-\alpha}).$$

For $G = GL_r$ identify these characters with the Schur functions

$$s_\lambda(x) = \frac{\det(x_i^{\lambda_j + r - j})}{\det(x_i^{r - j})},$$

and derive the tableaux expansion, Cauchy identity, and Littlewood–Richardson tensor-product rule.

Exercise 4.5 (Flavor multiplets and the Gell-Mann–Okubo prediction). Use the $SU(3)$ Weyl character $s_{(2,1)}$ to recover the eight weights of the adjoint representation with their multiplicities. Let isospin $SU(2)$ and hypercharge $U(1)$ be the evident subgroup and show that the octet decomposes as two isodoublets, one isotriplet, and one isosinglet. Assume the leading symmetry-breaking mass operator is an isoscalar in $1 \oplus 8$. Apply the Wigner–Eckart theorem and eliminate its three reduced parameters to derive

$$2(M_N + M_\Xi) = 3M_\Lambda + M_\Sigma.$$

Explain which four observed baryon centroid masses test the relation and separate the exact representation-theoretic multiplicities from the first-order symmetry-breaking assumption.

5 Coadjoint Orbits

Exercise 5.1 (Hamiltonian group actions and moment maps). Define smooth manifolds by compatible smooth atlases, smooth maps by their chart representatives, and tangent vectors intrinsically as derivations of germs. Construct tangent and cotangent bundles, differential forms, pullback, wedge product, and the exterior derivative, and prove their independence of charts. Define a Lie group as a group object in smooth manifolds and identify its Lie algebra with its left-invariant vector fields. Define symplectic manifolds and Hamiltonian vector fields by $\iota_{X_f} \omega = df$, and the Poisson bracket $\{f, g\} = \omega(X_f, X_g)$. For a Lie-group action define an equivariant moment map $\mu : M \rightarrow \mathfrak{g}^*$ by $d\langle \mu, \xi \rangle = \iota_{\xi_M} \omega$. Prove Noether’s theorem, the Poisson relation for moment-map components, and functoriality under products.

Exercise 5.2 (Kirillov–Kostant–Souriau orbits). For $\lambda \in \mathfrak{g}^*$ identify $T_\lambda(G\lambda)$ with $\mathfrak{g}/\mathfrak{g}_\lambda$ and define

$$\omega_\lambda(\text{ad}_\xi^* \lambda, \text{ad}_\eta^* \lambda) = \langle \lambda, [\xi, \eta] \rangle.$$

Prove that this is well defined, closed, nondegenerate, and has moment map the orbit inclusion. Identify the $SU(2)$ orbits with spheres, compute their symplectic areas intrinsically, and derive the Lie–Poisson equations.

Exercise 5.3 (Symplectic reduction and orbit quantization). For a regular value a of a moment map, construct $M//_a G = \mu^{-1}(a)/G_a$ and prove that its two-form is characterized by pullback. Define a prequantum Hermitian line bundle by a unitary connection of curvature $-i\omega/\hbar$. Prove the integrality condition and, for an $SU(2)$ orbit of area $2\pi\hbar n$, realize its quantization as the right-weight- n functions on $SU(2)$ annihilated by the positive-root operator. Use Peter–Weyl to identify this space with $\text{Sym}^n(\mathbb{C}^2)^\vee$. Derive quantization commutes with reduction for tensor products of these rank-one orbits and recover the triangle rule.

Exercise 5.4 (Spin coherent states and their kernels). Let J_1, J_2, J_3 be the infinitesimal $SU(2)$ generators normalized by $[J_a, J_b] = i\hbar\epsilon_{abc}J_c$. For $j = n/2$, define the normalized spin coherent state $|x\rangle$ as the image in $\text{Sym}^n(\mathbb{C}^2)$ of a unit vector projecting to $x \in S^2$. Prove

$$|\langle x | y \rangle|^2 = \left(\frac{1 + x \cdot y}{2} \right)^{2j}, \quad \frac{2j+1}{4\pi} \int_{S^2} |x\rangle\langle x| d\Omega = 1.$$

Expand a rotated coherent state into the J_3 weight lines and show that its probabilities are the binomial kernel

$$p_m(\theta) = \binom{2j}{j+m} \cos^{2(j+m)} \frac{\theta}{2} \sin^{2(j-m)} \frac{\theta}{2}.$$

Define the Krawtchouk polynomials by

$$\sum_{n=0}^N \binom{N}{n} K_n(x; p, N) t^n = \left(1 - \frac{1-p}{p} t \right)^x (1+t)^{N-x}.$$

Identify the associated rotation coefficients with these polynomials at $p = \sin^2(\theta/2)$ and derive their binomial orthogonality.

Exercise 5.5 (Spin precession and Stern–Gerlach intensities). For these infinitesimal generators let the Zeeman Hamiltonian be $H = -\gamma \mathbf{B} \cdot \mathbf{J}$. Solve the Hamiltonian orbit equation and the quantum evolution to obtain precession frequency $\omega_L = |\gamma| |\mathbf{B}|$. If a beam polarized along x is measured along an axis making angle θ with x , prove that the $2j+1$ separated intensities are precisely $p_m(\theta)$ above. Thus the coherent-state/Krawtchouk kernel predicts both the line count and every relative Stern–Gerlach intensity.

6 Heisenberg Group

Exercise 6.1 (Central extensions of phase space). Let (W, ω) be a finite-dimensional real symplectic vector space. Define the Heisenberg group

$$\text{Heis}(W) = W \times \mathbb{R}, \quad (w, s)(w', s') = (w + w', s + s' + \tfrac{1}{2}\omega(w, w')).$$

Prove associativity, identify its center and commutator, and show that its Lie algebra satisfies $[(w, s), (w', s')] = (0, \omega(w, w'))$. Prove that every continuous central extension of the additive group W with this commutator is equivalent to this one after changing its section by a continuous coboundary.

Exercise 6.2 (Weyl relations and the Schrödinger representation). For a real finite-dimensional vector space Q , put $W = Q \oplus Q^*$ and $\omega((q, p), (q', p')) = p(q') - p'(q)$. On $L^2(Q)$ with Lebesgue measure fixed by a density, define intrinsically

$$(T_q f)(x) = f(x - q), \quad (M_p f)(x) = e^{ip(x)/\hbar} f(x),$$

and $W(q, p) = e^{-ip(q)/(2\hbar)} M_p T_q$. Prove

$$W(w)W(w') = e^{i\omega(w, w')/(2\hbar)} W(w + w').$$

Differentiate on the Schwartz core to obtain $[\hat{p}, \hat{q}] = -i\hbar, p(q)$ and prove the Robertson uncertainty inequality directly from positivity of $\|(A - \langle A \rangle)x + it(B - \langle B \rangle)x\|^2$.

Exercise 6.3 (Stone–von Neumann decomposition). Call a unitary representation of $\text{Heis}(W)$ regular if its one-parameter subgroups are strongly continuous and assume its center acts by $e^{is/\hbar}$. For a Lagrangian subspace $Q^* \subset W$, construct the projection-valued measure obtained from its commuting translations. Prove the imprimitivity relation for the complementary translations, identify its spectral multiplicity as constant, and construct a unitary intertwiner with a Hilbert direct sum of Schrödinger representations. Deduce that every irreducible regular representation with this central character is unitarily equivalent to the Schrödinger representation.

Exercise 6.4 (Self-adjoint generators and Hermite functions). Prove Stone's theorem by constructing the spectral measure of a strongly continuous one-parameter unitary group, and define its self-adjoint generator. For $Q = \mathbb{R}$, take the closures of $Xf = xf$ and $Pf = -i\hbar f'$ from the Schwartz core. For

$$H = \frac{P^2}{2m} + \frac{m\omega^2 X^2}{2}, \quad a = \frac{m\omega X + iP}{\sqrt{2m\hbar\omega}},$$

prove $[a, a^*] = 1$ and $H = \hbar\omega(a^*a + 1/2)$. Define the Hermite polynomials by $H_n(u) = (-1)^n e^{u^2} (d/du)^n e^{-u^2}$ and the normalized Hermite functions h_n . Derive their differential equation, recurrence, orthogonality, Mehler kernel, and completeness without calling them a basis.

Exercise 6.5 (Oscillator spectroscopy). Use the Hermite decomposition to prove $E_n = \hbar\omega(n + 1/2)$. Couple the oscillator to a weak electric field by $V(t) = -qE_0 X \cos(\Omega t)$ and derive from the ladder relations that only $n \leftrightarrow n \pm 1$ transitions occur, with

$$|\langle n+1 | X | n \rangle|^2 = \frac{\hbar}{2m\omega}(n+1).$$

For a thermal state compute the Green-function absorption profile and show that its lines are centered at $\Omega = \omega$, have the resolution width from Hilbert Space, and have intensities in the ratios prescribed above. These positions and ratios are the spectroscopic prediction.

7 Unitary Ensembles

Exercise 7.1 (Unitary symmetry and spectral measures). On the real inner-product space of Hermitian endomorphisms of a finite-dimensional Hermitian space W , let $U(W)$ act by conjugation. For a real confining function $\mathcal{V}$, prove that

$$d\mathbb{P}(H) = Z^{-1} e^{-\text{Tr}(\mathcal{V}(H))} dH$$

is invariant whenever it is integrable. Disintegrate this measure along conjugacy orbits and identify the unordered eigenvalue map without choosing eigenvectors.

Exercise 7.2 (Weyl integration and eigenvalue repulsion). Compute the normal Jacobian of the conjugation orbit through a simple-spectrum Hermitian operator and prove the Weyl integration formula

$$\int F(H) dH = C_n \int_{\mathbb{R}^n} F(\lambda) \prod_{i < j} (\lambda_i - \lambda_j)^2 \prod_i d\lambda_i$$

for conjugation-invariant F . Determine C_n from a Gaussian integral. Show that the squared Vandermonde forces quadratic repulsion at collisions and rewrite the eigenvalue density as a determinant using monic polynomials orthogonal for $e^{-\mathcal{V}(x)} dx$.

Exercise 7.3 (Orthogonal-polynomial and Christoffel–Darboux kernels). Let p_k be the monic orthogonal polynomials for $e^{-\mathcal{V}(x)} dx$, assumed positive with all moments, and let $h_k = \|p_k\|^2$. Derive the three-term recurrence and prove the Christoffel–Darboux identity

$$\sum_{k=0}^{n-1} \frac{p_k(x)p_k(y)}{h_k} = \frac{p_n(x)p_{n-1}(y) - p_{n-1}(x)p_n(y)}{h_{n-1}(x-y)}.$$

For $\phi_k = p_k e^{-\mathcal{V}/2} / \sqrt{h_k}$ define $K_n(x, y) = \sum_{k \leq n} \phi_k(x)\phi_k(y)$. Apply the Andreief determinant identity to prove that every r -point eigenvalue correlation is $\det(K_n(x_i, x_j))_{i,j \leq r}$ and that K_n is an orthogonal-projection kernel.

Exercise 7.4 (Hermite, sine, and Airy kernels). For $\mathcal{V}(x) = x^2$, identify p_k with the Hermite polynomials already developed for the oscillator and compute their normalization. Use steepest descent in their contour representation to derive the bulk and soft-edge limits

$$\begin{aligned} \frac{1}{\rho_n(E)} K_n \left(E + \frac{u}{\rho_n(E)}, E + \frac{v}{\rho_n(E)} \right) &\longrightarrow \frac{\sin \pi(u-v)}{\pi(u-v)}, \\ \frac{1}{c_n} K_n \left(a_n + \frac{u}{c_n}, a_n + \frac{v}{c_n} \right) &\longrightarrow K_{\text{Ai}}(u, v) = \frac{\text{Ai}(u) \text{Ai}'(v) - \text{Ai}'(u) \text{Ai}(v)}{u-v}. \end{aligned}$$

Define $\text{Ai}(u) = (2\pi i)^{-1} \int_{\mathcal{C}} e^{t^3/3 - ut} dt$ on the standard Airy contour and derive both real asymptotic regimes and its differential equation.

Exercise 7.5 (Fredholm determinants and quantum-chaotic spectra). Define the singular values of a compact operator from the spectral theorem for $|A| = (A^*A)^{1/2}$ and call A trace class when their sum is finite. For positive trace-class B , define $\text{Tr}(B) = \sum_{\lambda > 0} \lambda \text{rank } P_{\{\lambda\}}$ and extend the trace to every trace-class operator through its decomposition into the real and imaginary parts and their positive and negative parts. Prove that the result is independent of the decomposition, that $|\text{Tr}(A)| \leq \text{Tr}|A|$, and that $\text{Tr}(AB) = \text{Tr}(BA)$ whenever one factor is trace class and the other bounded. Construct the Hilbert exterior powers $\wedge^r \mathcal{H}$ as the ranges of the antisymmetrizing projections on Hilbert tensor powers and show that $\wedge^r A$ is trace class. With no positivity or self-adjointness assumption, define

$$\det(1 + A) = \sum_{r \geq 0} \text{Tr}(\wedge^r A)$$

and prove absolute convergence, multiplicativity, and trace-norm continuity. Define a point process as a random locally finite counting measure and its correlation measures by factorial moments. Call it determinantal with kernel K when its correlation densities are $\det(K(x_i, x_j))_{i,j \leq r}$. Derive its generating functional and prove the gap formula $\mathbb{P}(N_D = 0) = \det(1 - 1_D K 1_D)$. In the sine-kernel limit derive $R_2(s) = 1 - (\sin \pi s / (\pi s))^2 = \pi^2 s^2 / 3 + O(s^4)$; at the edge define $F_2(s) = \det(1 - K_{\text{Ai}})_{L^2(s, \infty)}$. Define the Hastings–McLeod solution by

$$q''(s) = sq(s) + 2q(s)^3, \quad q(s) \sim \text{Ai}(s) \quad (s \rightarrow +\infty),$$

prove existence and uniqueness from its Volterra equation and resolvent estimates, and derive $F_2(s) = \exp(-\int_s^\infty (x-s)q(x)^2 dx)$. For a quantum system whose unfolded spectrum has no antiunitary symmetry, formulate the predictions of quadratic level repulsion, the sine-kernel two-point curve, and the F_2 largest-level law, separating the random-matrix modeling hypothesis from the measured spacings.

8 Poincaré Group

Exercise 8.1 (Minkowski isometries and their Casimirs). Let Minkowski spacetime be an oriented, time-oriented affine space modeled on (M, g) with signature $(-, +, +, +)$. Define $\mathcal{P}^+ = M \rtimes SO^+(M, g)$, derive its Lie algebra, and construct its metric volume form and Hodge star, then construct its two Casimirs from momentum P and the Pauli–Lubanski covector $W = *(P^\flat \wedge J)$, fixing this normalization. Prove $P^2 = -m^2 c^2$ and $W^2 = -m^2 c^2 \hbar^2 j(j+1)$ on a massive spin- j irreducible representation.

Exercise 8.2 (Wigner classification by little groups). For a transitive action of a locally compact group G on G/H , define the representation induced from a unitary H -representation as square-integrable equivariant sections of the associated bundle. Prove the imprimitivity theorem by disintegrating the translation spectral measure. Apply it to the mass shells in M^* : identify the massive little group with $SU(2)$ after passing to the double cover, the massless little group with $\text{Spin}(2) \ltimes \mathbb{R}^2$, where $\text{Spin}(2) = U(1) \rightarrow SO(2)$ is the angle-doubling cover, and derive the massive spin and massless helicity families. State explicitly the exclusion of continuous-spin representations when the translational part of the massless little group acts trivially.

Exercise 8.3 (Klein–Gordon Bessel functions). Put $\mu = mc/\hbar$ and $\square = -c^{-2}\partial_t^2 + \Delta$. Derive $(\square - \mu^2)\phi = 0$ from the massive Casimir. Define $\Gamma(s) = \int_0^\infty t^{s-1} e^{-t} dt$ for $\Re s > 0$, derive its functional equation and meromorphic continuation, and define

$$J_\nu(z) = \sum_{k \geq 0} \frac{(-1)^k (z/2)^{2k+\nu}}{k! \Gamma(k+\nu+1)}, \quad K_\nu(z) = \int_0^\infty e^{-z \cosh u} \cosh(\nu u) du$$

in their stated domains and continue by their differential equation. Use the Fourier decomposition of the positive mass shell to derive, at equal time and spatial separation $r > 0$,

$$\Delta_+(0, r) = \frac{mc}{4\pi^2 \hbar r} K_1(\mu r).$$

Derive the small- and large-argument asymptotics needed below.

Exercise 8.4 (Relativistic correlation length). For a free massive scalar field, define the vacuum two-point observable by the positive-energy mass-shell integral and derive the preceding K_1 formula without choosing an inertial frame until evaluating the invariant spacelike distance. Prove

$$\Delta_+(x, y) \sim \frac{\sqrt{m}}{4\sqrt{2} \pi^{3/2} (\hbar/c)^{1/2} r^{3/2}} e^{-r/\lambda_C}, \quad \lambda_C = \frac{\hbar}{mc},$$

up to the normalization fixed in the previous exercise. Show that the slope of $\log(r^{3/2} \Delta_+)$ against invariant separation is $-1/\lambda_C$; this is a quantitative mass determination from spacelike correlation data.

Part III

Scattering Theory

9 One-Dimensional Scattering

Exercise 9.1 (Asymptotic translation symmetry and scattering channels). On $L^2(\mathbb{R})$ let

$$H_0 = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2}, \quad H = H_0 + V,$$

where V is real and $\int_{\mathbb{R}} (1 + |x|) |V(x)| dx < \infty$. Construct H from its closed semibounded quadratic form and prove that its absolutely continuous spectrum on $(0, \infty)$ has two asymptotic translation channels. Construct the wave operators

$$\Omega_{\pm} = \text{s-lim}_{t \rightarrow \pm\infty} e^{iHt/\hbar} e^{-iH_0 t/\hbar}$$

first on Fourier transforms of compactly supported functions away from zero, using the Duhamel formula and the one-dimensional dispersive estimate, and then by density. Prove isometry, intertwining, and completeness from the Volterra resolvent. Define $S = \Omega_+^* \Omega_-$ and prove that it commutes with H_0 . Show that time reversal makes the two-channel scattering matrix symmetric and that an even potential further decomposes it into parity channels.

Exercise 9.2 (Jost solutions and the scattering matrix). For $k > 0$ define the Jost solutions as the unique solutions of

$$-\frac{\hbar^2}{2m} f'' + Vf = \frac{\hbar^2 k^2}{2m} f, \quad f_{\pm}(k, x) e^{\mp i k x} \rightarrow 1 \quad (x \rightarrow \pm\infty),$$

and construct them from their Volterra integral equations. Use their Wronskians to define transmission and left and right reflection amplitudes, fixing the convention

$$f_+(k, x) = t(k)^{-1} e^{i k x} + t(k)^{-1} r_-(k) e^{-i k x} + o(1) \quad (x \rightarrow -\infty).$$

Prove that these coefficients form the fiber of S , derive unitarity and reciprocity, and meromorphically continue them to $\Im k > 0$. Prove that poles of $t(k)$ at $k = i\kappa$ are bound states of energy $-\hbar^2 \kappa^2 / (2m)$, derive their norming constants from the residues, and identify lower-half-plane continuation poles as resonances.

Exercise 9.3 (Pöschl–Teller hypergeometric functions). For

$$V(x) = -\frac{\hbar^2}{2ma^2} \lambda(\lambda + 1) \operatorname{sech}^2(x/a), \quad \lambda > 0,$$

put $z = (1 - \tanh(x/a))/2$ and reduce the Jost equation to the Gauss equation. Define

$${}_2F_1(A, B; C; z) = \sum_{n \geq 0} \frac{(A)_n (B)_n}{(C)_n n!} z^n$$

for $|z| < 1$, where $(A)_0 = 1$ and $(A)_n = A(A+1) \cdots (A+n-1)$. Derive its differential equation and Euler integral, and use the integral to prove its connection formula between $z = 0$ and $z = 1$. Express both Jost solutions through these functions and obtain, with $p = ka$,

$$t(p) = \frac{\Gamma(1 + \lambda - ip) \Gamma(-\lambda - ip)}{\Gamma(1 - ip) \Gamma(-ip)},$$

in the preceding asymptotic convention. Use the gamma reflection formula to prove

$$|t(p)|^2 = \frac{\sinh^2(\pi p)}{\sinh^2(\pi p) + \sin^2(\pi \lambda)}, \quad |r(p)|^2 = \frac{\sin^2(\pi \lambda)}{\sinh^2(\pi p) + \sin^2(\pi \lambda)}.$$

Exercise 9.4 (Reflectionless transmission spectroscopy). For integral $\lambda = N$, prove that the Pöschl–Teller potential is reflectionless at every positive energy and that its bound-state poles give

$$E_n = -\frac{\hbar^2}{2ma^2}(N - n)^2, \quad n = 0, \dots, N - 1.$$

For nonintegral λ , locate the bound-state poles with $n \in \mathbb{Z}_{\geq 0}$ and $n < \lambda$ and the continued resonance poles, and derive the threshold and high-energy asymptotics of the reflection probability. Formulate a one-dimensional matter-wave or microwave-guide experiment that measures transmitted and reflected flux as a function of k : the model predicts the displayed transmission curve, exact reflectionlessness at integral N , and the bound-state line positions. Separate calibration of a and the potential depth, finite preparation width, absorption, and truncation of the ideal sech^2 profile from these predicted observables.

10 Potential Scattering

Exercise 10.1 ($SO(3)$ symmetry and radial operators). On $L^2(\mathbb{R}^3)$ let $SO(3)$ act by pullback. For a central potential $V(r)$, prove that the Schrödinger operator commutes with this action. Use the spherical-harmonic decomposition from Compact Harmonic Analysis to obtain the intrinsic orthogonal sum over angular momentum and the radial operators

$$H_\ell = -\frac{\hbar^2}{2m} \frac{d^2}{dr^2} + \frac{\hbar^2 \ell(\ell + 1)}{2mr^2} + V(r)$$

on $L^2(\mathbb{R}_+, dr)$ with the regular boundary condition.

Exercise 10.2 (Partial-wave decomposition and phase shifts). For a short-range real potential, construct the regular radial solution by a Volterra equation and prove that for $k > 0$ it has the asymptotic form

$$u_\ell(k, r) = A_\ell(k) \sin(kr - \ell\pi/2 + \delta_\ell(k)) + o(1).$$

Define $S_\ell(k) = e^{2i\delta_\ell(k)}$, fix $\delta_\ell = 0$ for the free operator, and use the conserved Wronskian to prove $|S_\ell| = 1$. Derive

$$f(\theta) = \frac{1}{2ik} \sum_{\ell \geq 0} (2\ell + 1)(S_\ell - 1)P_\ell(\cos \theta), \quad \sigma_{\text{tot}} = \frac{4\pi}{k^2} \sum_{\ell \geq 0} (2\ell + 1) \sin^2 \delta_\ell.$$

Exercise 10.3 (Coulomb hypergeometric functions). For $V(r) = \kappa/r$ set $\rho = kr$ and $\eta = m\kappa/(\hbar^2 k)$. Define ${}_1F_1(a; c; z) = \sum_{n \geq 0} (a)_n z^n / ((c)_n n!)$. Define the regular Coulomb function by

$$F_\ell(\eta, \rho) = C_\ell(\eta) \rho^{\ell+1} e^{-i\rho} {}_1F_1(\ell + 1 - i\eta; 2\ell + 2; 2i\rho),$$

choosing $C_\ell(\eta) > 0$ so that its large- ρ sine amplitude is one. Derive the radial equation, the Euler integral and connection formula for ${}_1F_1$, and $\sigma_\ell = \arg \Gamma(\ell + 1 + i\eta)$ with $\sigma_\ell - \sigma_{\ell-1} = \arctan(\eta/\ell)$. Sum the resulting Legendre series by its generating function.

Exercise 10.4 (Rutherford scattering). Show that Coulomb scattering has amplitude, up to an unobservable constant phase,

$$f_C(\theta) = -\frac{\eta}{2k \sin^2(\theta/2)} \exp \left[-2i\eta \log \sin \frac{\theta}{2} + 2i\sigma_0 \right].$$

Hence derive

$$\frac{d\sigma}{d\Omega} = \left(\frac{Z_1 Z_2 e^2}{16\pi\epsilon_0 E} \right)^2 \csc^4 \frac{\theta}{2}$$

for repulsive charges with center-of-mass kinetic energy E . Identify beam energy, scattering angle, and detector solid angle as the measured quantities, and state the point-charge and nonrelativistic limitations of the prediction.

11 Schwarzschild Spacetime

Exercise 11.1 (Lorentzian geometry and spherical vacuum metrics). Define a Lorentzian metric, its Levi-Civita connection by the Koszul identity, curvature, Ricci tensor, and geodesics. Derive the Jacobi equation from a variation of geodesics. For a static spherically symmetric metric, use the area radius and proper-time normalization at infinity to solve $\text{Ric} = 0$, obtaining

$$g = -\left(1 - \frac{r_s}{r}\right) c^2 dt^2 + \left(1 - \frac{r_s}{r}\right)^{-1} dr^2 + r^2 d\Omega^2, \quad r_s = \frac{2GM}{c^2}.$$

Prove that its connected exterior isometry group is $\mathbb{R} \times SO(3)$.

Exercise 11.2 (Regge–Wheeler decomposition). Set $G = c = 1$ temporarily and define the tortoise coordinate by $dr_*/dr = (1 - 2M/r)^{-1}$. Separate the scalar wave equation using the $SO(3)$ decomposition and prove that $u_{\ell m} = r\phi_{\ell m}$ satisfies

$$\left(-\partial_t^2 + \partial_{r_*}^2 - V_\ell(r)\right) u_{\ell m} = 0, \quad V_\ell(r) = \left(1 - \frac{2M}{r}\right) \left(\frac{\ell(\ell+1)}{r^2} + \frac{2M}{r^3}\right).$$

Construct the conserved Wronskian and the ingoing/outgoing scattering coefficients, and prove their flux identity.

Exercise 11.3 (Confluent Heun functions and quasinormal modes). Define a confluent Heun function as the Frobenius solution at a regular singular point of

$$y'' + \left(\alpha + \frac{\beta+1}{z} + \frac{\gamma+1}{z-1}\right) y' + \left(\frac{\mu}{z} + \frac{\nu}{z-1}\right) y = 0,$$

with its parameters and initial exponent specified. Transform the separated Schwarzschild radial equation to this form and derive the three-term recurrence of its Frobenius coefficients. Define a quasinormal frequency as a zero of the connection coefficient multiplying the incoming solution at infinity for the solution ingoing at the horizon. Express that coefficient as the limit of the recurrence's continued fraction and prove its convergence under the minimal-solution condition.

Exercise 11.4 (Black-hole ringdown in the eikonal limit). For $\ell \rightarrow \infty$, locate the maximum of V_ℓ at the unstable photon orbit $r = 3M$. Match parabolic-cylinder solutions across the maximum and derive, for fixed overtone n ,

$$\frac{GM\omega_{\ell n}}{c^3} = \frac{\ell + 1/2 - i(n + 1/2)}{3\sqrt{3}} + O(\ell^{-1}).$$

Show that the predicted waveform has oscillation frequency $\Re\omega/(2\pi)$ and damping time $1/|\Im\omega|$. Explain how the confluent-Heun continued fraction supplies the finite- ℓ correction and why the displayed asymptotic value must not be substituted as an exact $\ell = 2$ prediction.

Part IV

Hyperbolic Geometry

12 $SL_2(\mathbb{R})$

Exercise 12.1 (Hyperbolic isometries and the Casimir). Let $SL_2(\mathbb{R})$ act on $\mathbb{H} = \{x + iy : y > 0\}$ by Möbius maps. Prove that it preserves $ds^2 = (dx^2 + dy^2)/y^2$, identify the stabilizer of i with $SO(2)$, and derive the Iwasawa and Cartan decompositions. With the nonnegative convention define

$$\Delta_{\mathbb{H}} = -y^2(\partial_x^2 + \partial_y^2)$$

and show that it is the action of the quadratic Casimir on right- $SO(2)$ invariant functions.

Exercise 12.2 (Principal, complementary, and discrete series). Construct the spherical principal series by unitary induction from the upper triangular subgroup, the complementary series by its invariant kernel, and the holomorphic and antiholomorphic discrete series on weighted Bergman spaces. Determine their Casimir and $SO(2)$ spectra, including the signs of the discrete-series weights. Prove irreducibility by cyclicity of a K -finite vector and classify the irreducible unitary representations by the possible lowest/highest weight or continuous Casimir parameter.

Exercise 12.3 ($SL_2(\mathbb{R})$ Plancherel decomposition). Construct the standard intertwining operator between opposite principal series, compute its scalar on every $SO(2)$ type from the raising and lowering relations, and determine its poles and residues. Shift its contour to recover the discrete-series summands and derive the group Plancherel measure, with both even and odd principal-series densities. Prove inversion first for compactly supported smooth K -finite functions and extend it to L^2 .

Exercise 12.4 (Legendre functions and the Mehler–Fock transform). With fixed branches, use the Gauss hypergeometric function to define

$$P_{\nu}^{\mu}(z) = \frac{1}{\Gamma(1-\mu)} \left(\frac{z+1}{z-1} \right)^{\mu/2} {}_2F_1 \left(-\nu, \nu+1; 1-\mu; \frac{1-z}{2} \right).$$

Derive the associated Legendre equation and show that the spherical function at spectral parameter λ is $P_{-1/2+i\lambda}(\cosh r)$. From the radial Sturm–Liouville problem prove the Mehler–Fock inversion and Plancherel formulas, including the density $\lambda \tanh(\pi\lambda)$. Derive the analogous fixed- $SO(2)$ -type transform with $P_{-1/2+i\lambda}^{-m}$.

Exercise 12.5 (Hyperbolic-drum spectra). Let a membrane occupy a geodesic disk of curvature radius a and hyperbolic radius R , with wave speed v and Dirichlet boundary. Separate its wave equation into $SO(2)$ types and prove that its resonances are

$$\omega_{m,k} = \frac{v}{a} \sqrt{\frac{1}{4} + \lambda_{m,k}^2}, \quad P_{-1/2+i\lambda_{m,k}}^{-|m|}(\cosh R) = 0.$$

Prove the $m \neq 0$ double degeneracy and the Weyl asymptotic from the Mehler–Fock spectral measure. The zero locations, degeneracies, and area coefficient are the three measurable predictions.

13 $SL_2(\mathbb{Z})$

Exercise 13.1 (Modular group and its fundamental domain). Define $SL_2(\mathbb{Z})$, its action on $\mathbb{H}$, and $S\tau = -1/\tau$, $T\tau = \tau + 1$. Prove the presentation $PSL_2(\mathbb{Z}) = \langle S, T \mid S^2 = (ST)^3 = 1 \rangle$ by reducing points to the standard fundamental domain. Compute its hyperbolic area and the cusp, elliptic-point, and orbifold stabilizer data.

Exercise 13.2 (Eisenstein series and the scattering coefficient). Construct $L^2(SL_2(\mathbb{Z}) \backslash \mathbb{H})$ with hyperbolic measure and the nonnegative Laplacian. Define the Eisenstein series $E(\tau, s) = \sum_{\Gamma_\infty \backslash \Gamma} (\Im \gamma \tau)^s$, where Γ_∞ is the stabilizer of the cusp at infinity, for $\Re s > 1$. Define the zeta function $\zeta(s) = \sum_{n \geq 1} n^{-s}$ there and its completion $\xi(s) = \pi^{-s/2} \Gamma(s/2) \zeta(s)$. Unfold the Eisenstein series, continue it using the theta Mellin transform, and prove that its constant term is

$$E(x + iy, s)_0 = y^s + \varphi(s) y^{1-s}, \quad \varphi(s) = \frac{\xi(2s-1)}{\xi(2s)}.$$

Derive its functional equation and $\varphi(s)\varphi(1-s) = 1$, including unitarity on $\Re s = 1/2$.

Exercise 13.3 (Automorphic spectral and trace decomposition). Develop the Friedrichs realization of the nonnegative Laplacian at the cusp. Truncate the Eisenstein series, prove the Maass–Selberg relation, and obtain the orthogonal decomposition into constants, cusp forms, and the continuous Eisenstein integral. Build the automorphic kernel from a compactly supported point-pair invariant, unfold it by conjugacy classes, and derive the Selberg trace identity with its identity, elliptic, parabolic, and hyperbolic terms. Recover the Weyl area term.

Exercise 13.4 (Modular forms, theta constants, and the eta function). Define modular forms by their weight, multiplier, and cusp growth. Starting from the theta transformation proved in Fourier Analysis, define $\vartheta_2, \vartheta_3, \vartheta_4$ by their shifted or signed lattice sums and derive all their S, T transformations without re-proving Poisson summation. Define $\eta(\tau) = q^{1/24} \prod_{n \geq 1} (1 - q^n)$ with $q = e^{2\pi i \tau}$, and derive its transformations from the Jacobi triple product. Construct E_4, E_6 , prove the valence formula, and show

$$\mathbb{C}[E_4, E_6] = \bigoplus_{k \geq 0} M_k, \quad \Delta = \eta^{24} = \frac{E_4^3 - E_6^2}{1728}.$$

Exercise 13.5 (Eisenstein scattering on the modular surface). Model a scalar waveguide by the modular surface with its cusp attached to a single asymptotic lead. At spectral value $s = \frac{1}{2} + ir$ with $r > 0$, so that the nonnegative Laplacian has value $\frac{1}{4} + r^2$, identify $y^{1/2-ir}$ and $y^{1/2+ir}$ as the two cusp waves. Orient the lead so that the y^s coefficient is the unit incoming amplitude and the y^{1-s} coefficient is outgoing, and prove that the reflection amplitude is $\varphi(\frac{1}{2} + ir)$. Write $\varphi(\frac{1}{2} + ir) = e^{2i\delta(r)}$ and, for wave speed v and $\omega(r) = v\sqrt{\frac{1}{4} + r^2}$, derive the Wigner delay

$$\tau_W(r) = 2 \frac{d\delta}{d\omega} = \frac{2\sqrt{\frac{1}{4} + r^2}}{vr} \frac{d\delta}{dr}.$$

Show that the meromorphically continued poles of φ are the scattering resonances. Thus the completed-zeta quotient predicts the reflected phase, time delay, and resonance locations; separate the constant-curvature, single-cusp, scalar-wave, and lead-matching assumptions from those measured quantities.

Part V

Conformal Field Theory

14 Virasoro Algebra

Exercise 14.1 (The Virasoro central extension). On the punctured complex line let $L_n = -z^{n+1}\partial_z$, $n \in \mathbb{Z}$, be the distinguished Fourier vector fields. Prove $[L_m, L_n] = (m - n)L_{m+n}$. Define

$$\omega(L_m, L_n) = \frac{m^3 - m}{12} \delta_{m+n, 0}$$

and prove directly that it is alternating and satisfies the Lie-algebra cocycle identity. Define $\text{Vir} = \bigoplus_{n \in \mathbb{Z}} \mathbb{C}L_n \oplus \mathbb{C}C$ by

$$[L_m, L_n] = (m - n)L_{m+n} + \omega(L_m, L_n)C, \quad [C, L_n] = 0,$$

and verify the involution $L_n^* = L_{-n}$, $C^* = C$. Recover the cocycle as

$$\omega(f\partial_z, g\partial_z) = \frac{1}{12} \text{Res}_{z=0}(f'''(z)g(z) dz),$$

compute its change under a new circle coordinate as a coboundary, and deduce that the central-extension class is coordinate-independent.

Exercise 14.2 (Positive-energy highest-weight decomposition). Call a unitary Virasoro representation positive-energy when C acts by a real scalar c , L_0 has spectrum bounded below with finite-dimensional eigenspaces, and the algebraic sum of those eigenspaces is dense. Use $[L_0, L_n] = -nL_n$ and the involution to prove that every irreducible such representation is generated by a vector v satisfying

$$L_0v = hv, \quad L_nv = 0 \ (n > 0), \quad Cv = cv,$$

and prove the orthogonal decomposition of an arbitrary positive-energy representation into irreducible highest-weight summands by induction over its energy levels. Define the Verma module $M(c, h)$ by this universal highest-weight property. Apply the PBW theorem to show that its level- N subspace is indexed by partitions of N ; write $p(N)$ for their number and put $p(N) = 0$ for $N < 0$. Derive

$$\text{ch } M(c, h) = \text{Tr}_{M(c, h)} q^{L_0 - c/24} = \frac{q^{h - (c-1)/24}}{\eta(\tau)}, \quad q = e^{2\pi i \tau}.$$

Exercise 14.3 (Shapovalov determinants and unitary representations). Give $M(c, h)$ the contravariant form determined by $\langle v, v \rangle = 1$ and $L_n^* = L_{-n}$. Compute its Gram determinants at levels one and two. Put

$$c = 1 - 6(b - b^{-1})^2, \quad h_{r,s}(b) = \frac{(r/b - sb)^2 - (b^{-1} - b)^2}{4}.$$

Using the PBW filtration, construct a singular vector when $h = h_{r,s}(b)$, compare the degree and leading coefficient of the level- N determinant as a polynomial in h , and prove

$$\det_N(c, h) = C_N \prod_{\substack{r, s \geq 1 \\ rs \leq N}} (h - h_{r,s}(b))^{p(N - rs)}$$

with $C_N > 0$ fixed by the ordered PBW monomials. Track the signs between consecutive zero curves and quotient the null spaces to prove that irreducible positive-energy modules are unitary for $c \geq 1$, $h \geq 0$, and, below $c = 1$, precisely at

$$c_m = 1 - \frac{6}{m(m+1)}, \quad h_{r,s}^{(m)} = \frac{((m+1)r - ms)^2 - 1}{4m(m+1)}$$

with $m \geq 2$, $1 \leq r \leq m-1$, $1 \leq s \leq m$, modulo $(r, s) \sim (m-r, m+1-s)$.

Exercise 14.4 (Minimal-model characters). For $K \in \mathbb{N}$ and $\mu \in \mathbb{Z}$ define the level theta series

$$\Theta_{\mu,K}(\tau) = \sum_{n \in \mathbb{Z}} q^{(2Kn+\mu)^2/(4K)}.$$

Determine the inclusions among the singular Verma submodules at $(c_m, h_{r,s}^{(m)})$, resolve the irreducible quotient by their alternating chain, and take its Euler character to prove

$$\chi_{r,s}^{(m)}(\tau) = \frac{\Theta_{(m+1)r-ms, m(m+1)}(\tau) - \Theta_{(m+1)r+ms, m(m+1)}(\tau)}{\eta(\tau)}.$$

Use the Poisson summation formula to derive the finite S - and T -matrices of these characters. At $m = 3$ identify the three weights $0, 1/2, 1/16$ and prove

$$\chi_0 = \frac{1}{2} \left(\sqrt{\frac{\vartheta_3}{\eta}} + \sqrt{\frac{\vartheta_4}{\eta}} \right), \quad \chi_{1/2} = \frac{1}{2} \left(\sqrt{\frac{\vartheta_3}{\eta}} - \sqrt{\frac{\vartheta_4}{\eta}} \right), \quad \chi_{1/16} = \frac{1}{\sqrt{2}} \sqrt{\frac{\vartheta_2}{\eta}},$$

fixing the square roots by their positive q -expansions.

Exercise 14.5 (Virasoro characters and monstrous moonshine). Define a vertex operator algebra as a positive-energy Virasoro representation with vacuum $\mathbf{1}$, translation operator D , conformal vector, and state-field map $Y(a, z) = \sum_{n \in \mathbb{Z}} a(n) z^{-n-1}$ satisfying

$$Y(\mathbf{1}, z) = 1, \quad Y(a, z)\mathbf{1} \in a + zV[[z]], \quad [D, Y(a, z)] = \partial_z Y(a, z),$$

and $(z-w)^N[Y(a, z), Y(b, w)] = 0$ for sufficiently large N , with the conformal field equal to $T(z)$. Derive the mode Jacobi identity and prove that the graded dimension is a Virasoro character. Define $j = E_4^3/\Delta$ and use the valence formula to show that every modular function holomorphic away from the cusp is a polynomial in j . Compute

$$J(\tau) = j(\tau) - 744 = q^{-1} + 196884q + 21493760q^2 + \dots$$

Write $J(q) = \sum_{n \geq -1} c(n)q^n$. Construct the graded traces of the moonshine vertex operator algebra and prove its Borcherds denominator identity

$$p^{-1} \prod_{m>0, n \in \mathbb{Z}} (1 - p^m q^n)^{c(mn)} = J(p) - J(q).$$

Extract the denominator recursion and use the vacuum normalization and first graded pieces to prove that its identity character is J . Verify $196884 = 1 + 196883$ and $21493760 = 1 + 196883 + 21296876$ as decompositions into irreducible Monster dimensions.

Exercise 14.6 (Conformal spectra and the Cardy prediction). Define a unitary two-dimensional conformal theory on a circle of circumference L by commuting positive-energy Virasoro actions with central charges $c, \bar{c}$ and weights $h, \bar{h}$, and use units $\hbar = 1$ in conformal finite-size formulas. Define $T(z) = \sum_n L_n z^{-n-2}$ and transform it from the plane to the cylinder, deriving the Schwarzian contribution from the Virasoro cocycle. Obtain the Hamiltonian and momentum

$$H = \frac{2\pi v}{L} \left(L_0 + \bar{L}_0 - \frac{c + \bar{c}}{24} \right), \quad P = \frac{2\pi}{L} (L_0 - \bar{L}_0).$$

For a finite character family and nonnegative integral multiplicity matrix M , prove that the torus trace $Z = \sum_{i,j} M_{ij} \chi_i \bar{\chi}_j$ is modular invariant exactly when M commutes with its S, T matrices; identify the integral-spin and $c - \bar{c} \in 24\mathbb{Z}$ conditions imposed by T . If $h_{\min}$ is the least chiral weight coupled to the vacuum, define $c_{\text{eff}} = c - 24h_{\min}$. Apply the S transformation and a saddle-point estimate to prove

$$\log \rho(\Delta) = 2\pi \sqrt{\frac{c_{\text{eff}} \Delta}{6}} + o(\sqrt{\Delta}).$$

Thus predict both the finite-size ground-energy shift $-\pi v(c + \bar{c})/(12L)$ and the high-energy level-counting rate, specifying v, L , and the measured energies.

15 Ising Model

Exercise 15.1 ($\mathbb{Z}_2$ spin symmetry and the partition function). For a finite graph $\Gamma = (V, E)$ and spins $\sigma : V \rightarrow \{\pm 1\}$ define

$$Z_\Gamma(K, h) = \sum_{\sigma} \exp \left(K \sum_{ij \in E} \sigma_i \sigma_j + h \sum_{i \in V} \sigma_i \right).$$

Identify the global $\mathbb{Z}_2$ action at $h = 0$. Expand each edge factor into even and odd parts and prove the high-temperature loop expansion. Expand about the two ordered configurations to obtain the dual domain-wall expansion. On the square lattice derive Kramers–Wannier duality $\sinh(2K) \sinh(2K^*) = 1$, including all finite-torus boundary sectors.

Exercise 15.2 (Transfer-matrix decomposition). For an $M \times N$ periodic square lattice, construct the row Hilbert space as the tensor product of the two-dimensional spin object and define the positive transfer operator $T_M(K)$. Prove $Z_{M,N} = \text{Tr}(T_M^N)$ and that T_M commutes with global spin reversal and translation. Introduce the Clifford algebra with the fixed convention $\{\gamma_a, \gamma_b\} = 2\delta_{ab}$, construct the Jordan–Wigner operators, and reduce T_M in each spin structure to commuting two-dimensional momentum blocks. Derive their rapidities

$$\cosh \gamma(k) = \cosh(2K^*) \cosh(2K) - \sinh(2K^*) \sinh(2K) \cos k.$$

Exercise 15.3 (Onsager free energy). Put $\beta = (k_B T)^{-1}$. Take the thermodynamic limit of the largest transfer eigenvalue and derive the isotropic free energy

$$-\beta f(K) = \frac{1}{2} \log(2 \sinh 2K) + \frac{1}{2\pi} \int_0^\pi \text{arcosh}(\cosh 2K \coth 2K - \cos \theta) d\theta.$$

Locate its only nonanalytic point at $K_c = \frac{1}{2} \log(1 + \sqrt{2})$ and derive the logarithmic heat-capacity singularity.

Exercise 15.4 (Toeplitz determinants and spontaneous magnetization). Express a long-distance spin correlation as a Toeplitz determinant. Derive the scalar symbol by the Clifford decomposition, factor it into inside and outside analytic parts, and prove the strong Szegő limit needed for its ordered phase. Obtain

$$M(K) = \lim_{r \rightarrow \infty} \langle \sigma_0 \sigma_r \rangle^{1/2} = \left(1 - \sinh^{-4}(2K)\right)^{1/8}, \quad K > K_c.$$

Exercise 15.5 (Virasoro characters of the Ising transfer matrix). Use the convention $q = e^{2\pi i \tau}$ from $SL_2(\mathbb{Z})$. Take the scaling limit of the four transfer-matrix spin structures and factor their fermionic mode products by the Jacobi triple product. Identify the resulting three functions with the $c = 1/2$ Virasoro characters $\chi_0, \chi_{1/2}, \chi_{1/16}$. Match the lattice vacuum, energy, and spin sectors to the weights $0, 1/2, 1/16$, derive their coefficient multiplicities, and use their S, T matrices to construct the modular-invariant torus partition function $|\chi_0|^2 + |\chi_{1/2}|^2 + |\chi_{1/16}|^2$.

Exercise 15.6 (Ising criticality). Restore $K = J/(k_B T)$. Predict

$$k_B T_c = \frac{2J}{\log(1 + \sqrt{2})}, \quad M(T) \sim C(T_c - T)^{1/8}.$$

On a cylinder of circumference L , use the three Ising characters and the conformal Hamiltonian to derive the universal ground-energy correction $-\pi v_s/(12L)$ and excitation gaps $(2\pi v_s/L)(h + \bar{h} + n + \bar{n})$. State separately the square-lattice, nearest-neighbor, zero-field assumptions; the measured temperature, magnetization, and gaps; and the nonuniversal calibrations J and v_s .

Part VI

Quantum Algebra

16 Affine Kac–Moody Algebras

Exercise 16.1 (Central extensions of loop algebras). Let $\mathfrak{g}$ be a complex simple Lie algebra and normalize its invariant form $(\ , \)$ so that long roots have squared length two. Define the polynomial loop algebra $L\mathfrak{g} = \mathfrak{g} \otimes \mathbb{C}[z, z^{-1}]$ and

$$\hat{\mathfrak{g}} = L\mathfrak{g} \oplus \mathbb{C}K \oplus \mathbb{C}d$$

by

$$[x \otimes z^m, y \otimes z^n] = [x, y] \otimes z^{m+n} + m\delta_{m+n,0}(x, y)K, \quad [d, x \otimes z^n] = nx \otimes z^n.$$

Obtain the central term from the residue cocycle $\text{Res}_{z=0}(\partial_z x(z), y(z)) dz$ and prove the Jacobi identity. For the simply connected compact real form construct the corresponding $U(1)$ -central extension of its smooth loop group at an integer level k by exponentiating the cocycle. If ϑ is the left Maurer–Cartan form, define the Cartan three-form as the scalar multiple of $(\vartheta, [\vartheta, \vartheta])$ that integrates to one on every long-root $SU(2)$ subgroup, and prove independence of the filling disk from its integral cohomology class.

Exercise 16.2 (Integrable highest-weight decomposition). Extend a Cartan algebra of $\mathfrak{g}$ by K and d . Define the null root by $\delta(d) = 1$ and by zero on the other Cartan summands. With highest root

θ , set $\alpha_0 = \delta - \theta$ and $\alpha_0^\vee = K - \theta^\vee$, and define the affine Weyl group, level, and dominant affine weights. Apply the rank-one $\mathfrak{sl}_2$ strings for every affine simple root to prove that an irreducible highest-weight module is integrable at level $k \in \mathbb{N}$ exactly when its finite highest weight λ satisfies

$$\langle \lambda, \theta^\vee \rangle \leq k.$$

Call a unitary loop-group representation positive-energy when the rotation energy $-d$ has spectrum bounded below with finite-dimensional eigenspaces. Prove that every such representation decomposes orthogonally into these modules. For $\widehat{\mathfrak{sl}}_2$ identify the level- k summands $L_0, L_1, \dots, L_k$.

Exercise 16.3 (Sugawara Virasoro decomposition). Let $\mathcal{C}_{\mathfrak{g}} \in \mathfrak{g} \otimes \mathfrak{g}$ be the inverse tensor to the invariant form and define the dual Coxeter number $h^\vee$ by the scalar $2h^\vee$ through which the associated Casimir acts on $\mathfrak{g}$. Write $J_{x,n}$ for the action of $x \otimes z^n$ and set the linear map $\mathcal{N}_{p,q} : \mathfrak{g} \otimes \mathfrak{g} \rightarrow \text{End}(L)$ to send $x \otimes y$ to $J_{x,p}J_{y,q}$ for $p < 0$ and to $J_{y,q}J_{x,p}$ for $p \geq 0$. Define the Sugawara operators intrinsically by contraction with $\mathcal{C}_{\mathfrak{g}}$:

$$L_n = \frac{1}{2(k + h^\vee)} \sum_{m \in \mathbb{Z}} \mathcal{N}_{n-m,m}(\mathcal{C}_{\mathfrak{g}}).$$

Prove from the affine commutators that these operators satisfy the Virasoro relations and

$$c = \frac{k \dim \mathfrak{g}}{k + h^\vee}, \quad h_\lambda = \frac{(\lambda, \lambda + 2\rho)}{2(k + h^\vee)}.$$

For $\widehat{\mathfrak{sl}}_2$ derive $c = 3k/(k + 2)$ and $h_\ell = \ell(\ell + 2)/(4(k + 2))$.

Exercise 16.4 (Weyl–Kac characters). Put $y = e^{2\pi iz}$ and, without changing the modular nome convention, extend the level theta series by

$$\Theta_{m,K}(z, \tau) = \sum_{n \in \mathbb{Z}} q^{K(n+m/(2K))^2} y^{K(n+m/(2K))}.$$

Define $\chi_\ell^{(k)}(z, \tau) = \text{Tr}_{L_\ell}(q^{L_0 - c/24} y^{H/2})$, where H is the $\mathfrak{sl}_2$ coroot operator. Construct the affine BGG resolution by reflections in the walls of the fundamental alcove and take its Euler character to prove the Weyl–Kac formula. For $\widehat{\mathfrak{sl}}_2$ obtain

$$\chi_\ell^{(k)}(z, \tau) = \frac{\Theta_{\ell+1,k+2}(z, \tau) - \Theta_{-\ell-1,k+2}(z, \tau)}{\Theta_{1,2}(z, \tau) - \Theta_{-1,2}(z, \tau)}, \quad 0 \leq \ell \leq k.$$

Use Poisson summation to prove that their continuous values at $z = 0$ transform by

$$S_{\ell m} = \sqrt{\frac{2}{k+2}} \sin \frac{(\ell+1)(m+1)\pi}{k+2}, \quad T_{\ell \ell} = e^{2\pi i(h_\ell - c/24)}.$$

Derive orthogonality and completeness of this finite sine transform.

Exercise 16.5 (Knizhnik–Zamolodchikov conformal blocks). For finite-dimensional $\mathfrak{g}$ -modules $V_1, \dots, V_n$, let Ω_{ij} be the invariant Casimir contraction on the i th and j th factors. On the configuration space of distinct points define

$$\nabla_{KZ} = d - \frac{1}{k + h^\vee} \sum_{i < j} \Omega_{ij} d \log(z_i - z_j).$$

Prove the infinitesimal braid relations for the Ω_{ij} and hence flatness. For $k \geq 2$ and four fundamental $\mathfrak{sl}_2$ modules, decompose the invariant tensors through the two intermediate-spin channels, place three points at $0, 1, \infty$, and reduce the horizontal-section equation to the Gauss hypergeometric equation. Derive its Euler integral from the Gamma integral, use it to obtain the connection matrix between the 0 and 1 channels, and compute the two braid eigenvalues. Record the resulting braid representation by its eigenvalue ratio and channel-connection matrix.

Exercise 16.6 (Wess–Zumino–Witten finite-size spectroscopy). Define the diagonal $SU(2)_k$ Wess–Zumino–Witten state space as $\bigoplus_{\ell=0}^k L_\ell \otimes \overline{L}_\ell$. For a periodic critical system whose scaling limit has this state space, use the Sugawara Hamiltonian and the affine characters to derive

$$E_0(L) = e_\infty L - \frac{\pi v}{6L} \frac{3k}{k+2} + o(L^{-1}),$$

$$E_{\ell, N, \bar{N}}(L) - E_0(L) = \frac{2\pi v}{L} (2h_\ell + N + \bar{N}) + o(L^{-1}),$$

where e_∞ is the bulk ground-energy density and $N, \bar{N} \in \mathbb{Z}_{\geq 0}$ are descendant levels, with every degeneracy given by the corresponding character coefficients. For an antiferromagnetic spin-1/2 Heisenberg chain of N_s sites, set the lattice spacing to one, take $k = 1$, $L = N_s$, and $v = \pi J/2$ to predict the leading primary gap $\pi^2 J/(2N_s)$ and ground-energy correction $-\pi^2 J/(12N_s)$. Separate the WZW scaling hypothesis, periodic boundary condition, velocity calibration, and logarithmic finite-size corrections from the measured gaps and degeneracies.

17 Quantum Groups

Exercise 17.1 (The Hopf algebra $U_v(\mathfrak{sl}_2)$). For $v \in \mathbb{C}^\times$ not a root of unity, define $U_v(\mathfrak{sl}_2)$ by

$$KEK^{-1} = v^2 E, \quad KFK^{-1} = v^{-2} F, \quad [E, F] = \frac{K - K^{-1}}{v - v^{-1}},$$

with

$$\Delta K = K \otimes K, \quad \Delta E = E \otimes K + 1 \otimes E, \quad \Delta F = F \otimes 1 + K^{-1} \otimes F,$$

and construct its counit and antipode from the Hopf identities. Define $[n]_v = (v^n - v^{-n})/(v - v^{-1})$ and construct every finite-dimensional type-one irreducible V_n from a highest-weight vector. Prove the deformed Clebsch–Gordan rule $V_m \otimes V_n \simeq \bigoplus_{j=0}^{\min(m, n)} V_{m+n-2j}$.

Exercise 17.2 (Universal R -matrix). Write $K = v^H$ formally and, in the completion acting on type-one modules, define

$$\mathcal{R} = v^{H \otimes H/2} \sum_{r \geq 0} v^{r(r-1)/2} \frac{(v - v^{-1})^r}{[r]_v!} E^r \otimes F^r.$$

Verify directly on highest-weight modules that $\mathcal{R}\Delta(a) = \Delta^{\text{op}}(a)\mathcal{R}$ and

$$(\Delta \otimes \text{id})\mathcal{R} = \mathcal{R}_{13}\mathcal{R}_{23}, \quad (\text{id} \otimes \Delta)\mathcal{R} = \mathcal{R}_{13}\mathcal{R}_{12}.$$

Define the braid group B_n by generators σ_i with the far-commutativity and braid relations; deduce its actions on tensor powers and the quantum Yang–Baxter equation. On the two fundamental channels, where the sum terminates and has a regular root-of-unity value, specialize at $v = e^{\pi i/(k+2)}$. Prove that the ratio of its two braid eigenvalues and its recoupling matrix agree with the framing-independent Knizhnik–Zamolodchikov braid data.

Exercise 17.3 (Root-of-unity tilting modules and ribbon quotients). Construct the divided-power integral form of $U_v(\mathfrak{sl}_2)$ over $\mathbb{Z}[v, v^{-1}]$ and specialize it at $v = e^{\pi i/(k+2)}$. Define Weyl modules, dual Weyl modules, and tilting modules through filtrations by both. Use the antipode, evaluation, and coevaluation to define their rigid duals and prove closure of the tilting subcategory under tensor products and duals. Construct the pivotal element from the square of the antipode and use it to define the categorical trace. Define a morphism to be negligible when its categorical trace vanishes after every composable morphism, prove that negligible morphisms form a tensor ideal, and form the Karoubi-completed quotient by formally adjoining the images of idempotents. Show that it is semisimple with simple objects $V_0, \dots, V_k$, quantum dimensions

$$d_a = \frac{\sin((a+1)\pi/(k+2))}{\sin(\pi/(k+2))},$$

and truncated Clebsch–Gordan products. Construct the ribbon element from the specialized R -matrix, define its compatibility with the coproduct, antipode, and pivotal element, prove the ribbon identities in the quotient, and derive

$$\theta_a = \exp\left(\frac{\pi i a(a+2)}{2(k+2)}\right).$$

Exercise 17.4 (Quantum Peter–Weyl decomposition). Define the restricted Hopf dual $\mathcal{O}_v(SL_2)$ as the span of matrix coefficients. Prove its algebraic Peter–Weyl decomposition

$$\mathcal{O}_v(SL_2) = \bigoplus_{n \geq 0} V_n^\vee \otimes V_n$$

and derive Schur orthogonality for the invariant positive functional when $0 < v < 1$, after equipping it with the compact real-form involution and writing the resulting Hopf $*$ -algebra as $\mathcal{O}_v(SU(2))$.

Exercise 17.5 (Basic hypergeometric and dual q -Hahn functions). Put $Q = v^2$ and keep this base distinct from the modular nome. Define $(a; Q)_r$ and

$${}_r\phi_s\left(\begin{matrix} a_1, \dots, a_r \\ b_1, \dots, b_s \end{matrix}; Q, z\right) = \sum_{k \geq 0} \frac{(a_1, \dots, a_r; Q)_k}{(Q, b_1, \dots, b_s; Q)_k} \left((-1)^k Q^{k(k-1)/2}\right)^{1+s-r} z^k.$$

Derive the Q -binomial and Q -Gauss sums by finite telescoping and analytic continuation. Define the dual Q -Hahn polynomials by

$$R_n(Q^{-x} + \gamma\delta Q^{x+1}; \gamma, \delta, N \mid Q) = {}_3\phi_2\left(\begin{matrix} Q^{-n}, Q^{-x}, \gamma\delta Q^{x+1} \\ \gamma Q, Q^{-N} \end{matrix}; Q, Q\right).$$

Derive their difference equation and finite orthogonality. Solve the $U_v(\mathfrak{sl}_2)$ lowering recurrence and identify its normalized Clebsch–Gordan coefficients with these polynomials after matching the three integer labels and normalization. Define the terminating Q -Racah family by

$$\mathcal{R}_n(Q^{-x} + \delta Q^{x-N}) = {}_4\phi_3\left(\begin{matrix} Q^{-n}, \alpha\beta Q^{n+1}, Q^{-x}, \delta Q^{x-N} \\ \alpha Q, \beta\delta Q, Q^{-N} \end{matrix}; Q, Q\right)$$

and derive its finite orthogonality and recoupling identity from two associations of a fourfold tensor product.

Exercise 17.6 (Six-vertex transfer matrices and the XXZ chain). Write $v = e^\eta$ and define, with additive spectral parameter u ,

$$R(u) = \begin{pmatrix} \sinh(u + \eta) & 0 & 0 & 0 \\ 0 & \sinh u & \sinh \eta & 0 \\ 0 & \sinh \eta & \sinh u & 0 \\ 0 & 0 & 0 & \sinh(u + \eta) \end{pmatrix}.$$

Evaluate the finite universal R -matrix on $V_1 \otimes V_1$, prove that its checked form satisfies the Hecke quadratic relation, and derive the displayed spectral matrix by Baxterization. Prove its Yang–Baxter equation directly from the braid and Hecke relations. Form the periodic row transfer matrix, prove commutativity for different u , and take its logarithmic derivative at $u = 0$ to obtain, up to scalar and scale,

$$H_{XXZ} = \sum_{j=1}^N (\sigma_j^x \sigma_{j+1}^x + \sigma_j^y \sigma_{j+1}^y + \cosh \eta \sigma_j^z \sigma_{j+1}^z).$$

Develop the algebraic Bethe ansatz and derive the Bethe equations and energy in every fixed-magnetization sector.

Exercise 17.7 (XXZ multiplet spectroscopy). For an open $U_v(\mathfrak{sl}_2)$ -symmetric spin chain, decompose three adjacent spin sectors in the two recoupling orders. Express the change of channel through normalized dual Q -Hahn coefficients and its iterates through Q -Racah polynomials. For a weak probe transforming as V_2 , derive the allowed total-spin channels and their relative transition intensities as squares of these coefficients. For the periodic chain, use the Bethe roots to give the corresponding resonance frequencies. State boundary symmetry, anisotropy $\Delta = \cosh \eta$, polarization, temperature, and line-resolution assumptions separately from the predicted missing lines, intensity ratios, and frequencies.

18 Double Affine Hecke Algebras

Exercise 18.1 (Double-affine braid groups). For the type- A_{r-1} root system in $\{x \in \mathbb{R}^r : \sum x_i = 0\}$, construct its Weyl chambers, affine reflection hyperplanes, and alcoves from the root datum already developed. Identify the affine braid group with braids in an annulus. Show that allowing every strand to wind around both cycles of a torus gives the double-affine braid group, with finite braid generators T_i and commuting winding families X_j and Y_j . Derive its braid and winding relations directly from homotopies of torus braids. Impose $(T_i - t^{1/2})(T_i + t^{-1/2}) = 0$ and fix the total winding commutator to q to define $H_{q,t}(GL_r)$. Here q is a multiplicative difference parameter, not the earlier modular nome; when this finite Hecke action is compared with Quantum Groups use $t^{1/2} = v$, and otherwise keep t and v independent.

Exercise 18.2 (Polynomial representation and Cherednik operators). Let

$$\mathcal{P}_r = \mathbb{C}(q, t^{1/2})[x_1^{\pm 1}, \dots, x_r^{\pm 1}].$$

Let X_i be multiplication, let s_i exchange x_i, x_{i+1} , define $T_{q,x_i} f(x_1, \dots, x_r) = f(x_1, \dots, qx_i, \dots, x_r)$, and put

$$T_i = t^{1/2} s_i + (t^{1/2} - t^{-1/2}) \frac{s_i - 1}{x_i/x_{i+1} - 1}, \quad (\omega f)(x) = f(qx_r, x_1, \dots, x_{r-1}),$$

$$Y_i = T_i \cdots T_{r-1} \omega T_1^{-1} \cdots T_{i-1}^{-1}.$$

Verify preservation of Laurent polynomials, the Hecke and affine braid relations, and $[Y_i, Y_j] = 0$. Prove their triangular action on monomials in Bruhat-refined dominance order. Show on symmetric Laurent polynomials that

$$t^{(r-1)/2} \sum_i Y_i = D_{q,t}, \quad D_{q,t} = \sum_i \prod_{j \neq i} \frac{tx_i - x_j}{x_i - x_j} T_{q,x_i}.$$

Exercise 18.3 (Macdonald polynomials and their kernel). For a partition λ of length at most r , define P_λ as the monic symmetric polynomial triangular in monomial symmetric functions and satisfying

$$D_{q,t} P_\lambda = \left(\sum_{i=1}^r q^{\lambda_i} t^{r-i} \right) P_\lambda.$$

For $0 < q, t < 1$, prove symmetry of $D_{q,t}$ for the torus weight

$$\Delta_{q,t}(x) = \prod_{i \neq j} \frac{(x_i/x_j; q)_\infty}{(tx_i/x_j; q)_\infty}$$

and deduce orthogonality and completeness. In the stable symmetric-function ring define the power-sum pairing

$$\langle p_\lambda, p_\mu \rangle_{q,t} = \delta_{\lambda\mu} z_\lambda \prod_j \frac{1 - q^{\lambda_j}}{1 - t^{\lambda_j}}.$$

Determine the Macdonald norm from the adjoint Pieri operators, define Q_λ to be dual to P_λ for this pairing, and prove the Cauchy kernel

$$\sum_\lambda P_\lambda(a; q, t) Q_\lambda(b; q, t) = \prod_{i,j} \frac{(ta_i b_j; q)_\infty}{(a_i b_j; q)_\infty}.$$

Construct nonsymmetric joint Y -eigenfunctions by intertwiners and recover P_λ with the finite-Hecke symmetrizer.

Exercise 18.4 (Elliptic deformation and trigonometric degeneration). For $|p|, |q| < 1$ define

$$\theta(z; p) = (z; p)_\infty (p/z; p)_\infty, \quad \Gamma(z; p, q) = \prod_{j,k \geq 0} \frac{1 - p^{j+1} q^{k+1} / z}{1 - p^j q^k z}.$$

Prove their difference and reflection identities. Use the multiplicative theta addition formula to construct the rank-one elliptic difference-reflection operators and verify their dynamical braid relation. Derive the terminating Frenkel–Turaev sum by telescoping that relation. Take $p \rightarrow 0$ to obtain the Askey–Wilson operator and its ${}_4\phi_3$ eigenpolynomials; specialize its four boundary parameters to recover the rank-one Macdonald operator. Then take $q = e^{-\epsilon}$, $t = q^\theta$, $\epsilon \downarrow 0$, to obtain the Jacobi/Jack differential operator. Record every parameter map so that p is always the elliptic nome, q the difference step, and t the Hecke parameter.

Exercise 18.5 (Ruijsenaars–Schneider spectrum). Put $x_j = e^{i\theta_j}$ and regard the commuting symmetric polynomials in Y_i as quantum integrals of the compact trigonometric Ruijsenaars–Schneider system. With the positive Macdonald weight, construct their self-adjoint realization and prove

$$E_\lambda = \sum_{i=1}^r q^{\lambda_i} t^{r-i}$$

for the first normalized Hamiltonian. Derive the first Macdonald Pieri rule for multiplication by $x_1 + \cdots + x_r$ and hence the allowed transitions $\lambda \rightarrow \lambda + \square$ and their squared normalized intensities. In the $q = e^{-\epsilon}$, $t = q^g$ limit recover the Sutherland energies and Jack selection rule. Thus a periodic density probe predicts exact line positions, missing lines, and intensities; state the finite-particle, coupling, boundary, and spectral-resolution assumptions.

Part VII

Algebraic Geometry

19 Algebraic Curves

Exercise 19.1 (Affine and projective curves). For a commutative unital ring A , define prime ideals, localization, the Zariski space $\text{Spec } A$, and its structure sheaf by the rings A_f on distinguished opens. Prove that every stalk is local and define a locally ringed space, morphisms of locally ringed spaces, affine schemes, and schemes by affine covers. Define Noetherian rings, Krull dimension, and a regular Noetherian local ring by $\dim_{A/\mathfrak{m}}(\mathfrak{m}/\mathfrak{m}^2) = \dim A$. Construct Proj from homogeneous prime ideals and homogeneous localization. Define a projective curve over $\mathbb{C}$ to be a smooth connected one-dimensional closed subscheme of a projective space, with smoothness expressed by regular local rings. Construct the analytification of a smooth projective curve from its local rings and prove it is a compact Riemann surface. Define divisors by valuations of the function field and prove that principal divisors have degree zero by the argument principle.

Exercise 19.2 (Homology, periods, and de Rham's theorem). Define singular chains, boundary, homology, cochains, cup product, orientations, and the fundamental class of an oriented closed manifold. For the differential forms developed with Hamiltonian actions, define integration over smooth singular simplices. Prove Stokes's theorem and show that integration gives a cochain map. Construct a good cover, compare the singular and de Rham Čech double complexes with partitions of unity, and prove

$$H_{\text{dR}}^*(M) \simeq H^*(M; \mathbb{R})$$

as graded rings. Derive Mayer–Vietoris and Poincaré duality from the integration pairing. For a compact Riemann surface, prove that the intersection form on H_1 is unimodular and alternating and construct a symplectic marking. Define its genus g by $\text{rank } H_1 = 2g$ and identify this integer with its topological handle number.

Exercise 19.3 (Elliptic curves and translation symmetry). For a nonsingular cubic $E : y^2 = 4x^3 - g_2x - g_3$ with discriminant $g_2^3 - 27g_3^2 \neq 0$, construct the group law by divisors and prove the chord–tangent formula. Show that the holomorphic form dx/y has no zeros and that its period map identifies

$$E(\mathbb{C}) \simeq \mathbb{C}/(\mathbb{Z}\omega_1 + \mathbb{Z}\omega_2).$$

Construct the Weierstrass function from its normally convergent lattice sum, prove $\wp'^2 = 4\wp^3 - g_2\wp - g_3$, and show that $z \mapsto (\wp(z), \wp'(z))$ is the inverse uniformization. Derive the addition formula from translation on E .

Exercise 19.4 (Branched covers and Riemann–Hurwitz). Define a branched cover of compact curves and its ramification divisor. Define the canonical divisor as the divisor of any nonzero

meromorphic differential and prove that its divisor class is independent of that differential. Prove Riemann–Hurwitz by comparing canonical divisors. For the normalization of $y^2 = \prod_{j=1}^{2g+1} (x - e_j)$, include the chart at infinity and prove smoothness and genus g .

Exercise 19.5 (Abel–Jacobi decomposition). Construct a symplectic marking of first homology, normalize the holomorphic differentials, and define the Abel map and Jacobian. Prove Abel’s theorem and Jacobi inversion by the argument principle and the Riemann bilinear relations.

Exercise 19.6 (Elliptic integrals and Jacobi functions). For $0 < k < 1$ define

$$K(k) = \int_0^{\pi/2} \frac{d\theta}{\sqrt{1 - k^2 \sin^2 \theta}}, \quad E(k) = \int_0^{\pi/2} \sqrt{1 - k^2 \sin^2 \theta} d\theta.$$

Invert the Abel integral $u = \int_0^x dt / \sqrt{(1 - t^2)(1 - k^2 t^2)}$ on its elliptic curve to define $\operatorname{sn}(u; k)$ globally; define cn and dn by the algebraic relations with branches fixed at zero. Derive their differential equations, addition laws, periods, Fourier series, and the hypergeometric representation $K(k) = \frac{\pi}{2} {}_2F_1(1/2, 1/2; 1; k^2)$.

Exercise 19.7 (Nonlinear pendulum periods). For a pendulum of length L in uniform gravity g , use energy conservation and the elliptic Abel map to prove that oscillations of amplitude $0 < \theta_0 < \pi$ have period

$$T(\theta_0) = 4\sqrt{\frac{L}{g}} K\left(\sin \frac{\theta_0}{2}\right).$$

Expand the hypergeometric series to obtain

$$T = 2\pi\sqrt{\frac{L}{g}} \left(1 + \frac{\theta_0^2}{16} + \frac{11\theta_0^4}{3072} + O(\theta_0^6)\right)$$

and derive the logarithmic divergence at the separatrix. Identify L, g, θ_0 , and the measured return time, and distinguish ideal rigid, frictionless motion from experimental corrections.

20 Riemann–Roch

Exercise 20.1 (Line bundles, sections, and sheaf cohomology). Define a line bundle $p : L \rightarrow X$ and immediately define a section over $U \subset X$ as a map $s : U \rightarrow L$ with $p \circ s = \operatorname{id}_U$; write $\Gamma(U, L)$ for all such sections. Define the sheaf of sections, locally free and coherent sheaves, tensor product, dual, and pullback. Construct Čech cohomology from an open cover. Prove the affine vanishing theorem for quasi-coherent sheaves directly from localization, deduce that affine covers compute their sheaf cohomology, prove independence of the chosen affine cover by the refinement double complex, and derive the long exact sequence from a short exact sequence of sheaves. Identify divisors modulo principal divisors with the Picard group.

Exercise 20.2 (Serre duality and Riemann–Roch). For a smooth projective curve X , construct the residue pairing

$$H^1(X, L) \otimes H^0(X, K_X \otimes L^{-1}) \longrightarrow \mathbb{C}$$

from Čech representatives and prove it is perfect. Prove Riemann–Roch by induction on a divisor using the exact sequence at a point:

$$h^0(L) - h^1(L) = \deg L + 1 - g.$$

Define the holomorphic Euler characteristic $\chi(X, L) = \sum_q (-1)^q \dim H^q(X, L)$. Deduce $\deg K_X = 2g - 2$, the vanishing theorem for $\deg L > 2g - 2$, and the dimension of meromorphic functions with prescribed poles.

Exercise 20.3 (Dolbeault cohomology and Serre duality). Define a complex manifold by a holomorphic atlas and a holomorphic vector bundle by its locally free sheaf of sections. Decompose complex differential forms into bidegrees and define ∂ and $\bar{\partial}$. Prove the $\bar{\partial}$ -Poincaré lemma, construct the fine Dolbeault resolution, and identify coherent sheaf cohomology of a holomorphic bundle E with $H_{\bar{\partial}}^{0,q}(X, E)$. On a compact complex manifold with Hermitian metric, construct the formal adjoint and Dolbeault Laplacian from the metric volume form, establish elliptic regularity from local Fourier estimates and a partition of unity, and prove the harmonic-representative theorem. Use the Hodge star and integration pairing to prove the perfect duality

$$H^q(X, E) \otimes H^{d-q}(X, K_X \otimes E^\vee) \longrightarrow \mathbb{C}$$

for $d = \dim_{\mathbb{C}} X$, recovering the curve theorem as the case $d = 1$.

Exercise 20.4 (Bott–Borel–Weil for SL_2). Identify $SL_2(\mathbb{C})/B \simeq \mathbb{P}^1$ and compute

$$H^q(\mathbb{P}^1, \mathcal{O}(n)) = \begin{cases} \text{Sym}^n(\mathbb{C}^2)^\vee, & q = 0, n \geq 0, \\ \text{Sym}^{-n-2}(\mathbb{C}^2), & q = 1, n \leq -2, \\ 0, & \text{otherwise,} \end{cases}$$

using Riemann–Roch and Serre duality. Recover the shifted reflection $s \cdot n = -n - 2$.

Exercise 20.5 (Bott–Borel–Weil for reductive groups). For general reductive G , cover G/B by Bruhat cells, filter its Čech complex by Bruhat order, identify the associated graded terms with Verma modules, and prove that singular $\lambda + \rho$ gives zero while regular $\lambda + \rho$ gives

$$H^{\ell(w)}(G/B, L_\lambda) \simeq V_{w(\lambda+\rho)-\rho}^\vee$$

for the unique w carrying it to the dominant chamber.

Exercise 20.6 (Twisted differential operators on $\mathbb{P}^1$). Derive from the Möbius action on $\mathbb{P}^1$ the operators on $\Gamma(\mathbb{A}^1, \mathcal{O}(n))$

$$e = \partial_z, \quad h = -2z\partial_z + n, \quad f = -z^2\partial_z + nz,$$

and verify the $\mathfrak{sl}_2$ relations and invariance of polynomials of degree at most n . Define $\Omega = \frac{1}{2}h^2 + ef + fe$, compute its scalar, and define the resulting infinitesimal character. Construct the twisted differential-operator sheaf $\mathcal{D}_n$ by gluing the displayed chart operators.

Exercise 20.7 (Beilinson–Bernstein localization). For a weight λ , define $\mathcal{D}_\lambda$ by twisting differential operators with L_λ . Construct the order filtration, identify its associated graded with functions on $T^*(G/B)$, and use the moment map and the Koszul resolution of its fibers to prove

$$\Gamma(G/B, \mathcal{D}_\lambda) \simeq U(\mathfrak{g})/U(\mathfrak{g}) \ker \chi_\lambda.$$

For regular dominant λ , prove exactness and faithfulness of global sections by the Bruhat filtration and deduce the localization equivalence. Recover Borel–Weil from the localized highest-weight module.

Exercise 20.8 (Bergman kernels). Give X a Hermitian area form and L a positive Hermitian holomorphic line bundle. Define the orthogonal projection $L^2(X, L) \rightarrow H^0(X, L)$ by its intrinsic integral kernel $B_L(x, y)$. Prove its reproducing, Hermitian, and projection identities. Define Λ as contraction with the area form. Derive the extremal characterization $B_L(x, x) = \sup\{|s(x)|^2 : \|s\|_{L^2} = 1\}$ and, from rescaling the heat kernel of the Dolbeault Laplacian, prove for L^k

$$B_{L^k}(x, x) = \frac{k}{2\pi} i\Lambda F_L + O(1)$$

uniformly. On $\mathbb{P}^1$ and on a complex torus compute the kernel explicitly, reusing rather than redefining the earlier theta functions.

Exercise 20.9 (Landau-level degeneracy and density). Let a charged particle move on a compact Riemann surface in a magnetic line bundle L with $\frac{q}{2\pi\hbar} \int_X B = N \in \mathbb{Z}$. Factor its magnetic Laplacian through the Dolbeault operator and identify the lowest Landau level with $H^0(X, L)$. When $N > 2g - 2$, prove

$$\dim H^0(X, L) = N + 1 - g, \quad \rho(x) = B_L(x, x).$$

Thus predict both the exact number of lowest-level states and their local particle density; state flux quantization, curvature, boundary, interaction, and field-uniformity assumptions.

21 Finite-Gap Spectral Curves

Exercise 21.1 (KdV Lax flows). On periodic smooth functions define $L = -\partial_x^2 + u$ and $P = -4\partial_x^3 + 3(u\partial_x + \partial_x u)$. Compute the commutator and prove that $L_t = [P, L]$ is

$$u_t - 6uu_x + u_{xxx} = 0.$$

Construct the monodromy of $L\psi = E\psi$, prove that its discriminant is constant under the flow, and show that the periodic spectrum is preserved. Derive the commuting odd operator for a stationary higher KdV flow and the Burchall–Chaundy polynomial relation.

Exercise 21.2 (Hyperelliptic spectral curves). Normalize the projective closure of $y^2 = \prod_{j=0}^{2g} (E - E_j)$ and include its smooth chart at infinity. Choose real ordered branch points and construct the real ovals. Normalize the holomorphic differentials by a symplectic homology marking and prove from the bilinear relations that the period matrix is symmetric with positive imaginary part. Define

$$\Theta(z \mid \Omega) = \sum_{n \in \mathbb{Z}^g} e^{\pi i n^T \Omega n + 2\pi i n^T z}$$

and derive its quasi-periodicity. State and prove the symplectic change of marking formula and the invariance of the reconstructed potential below.

Exercise 21.3 (Baker–Akhiezer functions). At the point over $E = \infty$, put $k = \sqrt{E}$. Construct normalized second-kind differentials with principal parts $d(ik)$ and $d(4ik^3)$ and period vectors U, V . By Riemann–Roch prove uniqueness of the function with a prescribed nonspecial pole divisor and essential factor $e^{ikx + 4ik^3 t}$. Derive its theta formula and show directly that it solves $L\psi = E\psi$ and $\psi_t = P\psi$. Compare the coefficient at infinity to prove

$$u(x, t) = c - 2\partial_x^2 \log \Theta(Ux + Vt + Z \mid \Omega)$$

and hence the stated sign convention for KdV.

Exercise 21.4 (Bloch–Floquet decomposition). For a real periodic potential of period a , define the translation by a and decompose $L^2(\mathbb{R})$ over its $U(1)$ characters. Prove that $\Delta(E) = \text{Tr } M(E)$ satisfies $\Delta(E) = 2 \cos(ka)$ on a Bloch wave and that the spectrum is exactly $\{E : |\Delta(E)| \leq 2\}$. Show that simple branch points of the spectral curve are band edges and that closing all but g gaps is equivalent to the finite-genus polynomial relation above.

Exercise 21.5 (Lamé band-gap spectroscopy). For

$$L_k = -\frac{d^2}{dx^2} + 2k^2 \text{sn}^2(x; k)$$

of period $2K(k)$, use the Jacobi identities to prove that dn , cn , and sn have band edge energies $k^2, 1, 1 + k^2$. Prove

$$\text{spec}(L_k) = [k^2, 1] \cup [1 + k^2, \infty).$$

After restoring a spatial scale κ and factor $\hbar^2/(2m)$, predict the two band widths and the gap width $\hbar^2 \kappa^2 k^2/(2m)$. State how Bloch spectroscopy measures the three edge energies and distinguish the one-gap periodic-potential model from lattice, interaction, and finite-size corrections.

Part VIII

Algebraic Topology

22 Index Theory

Exercise 22.1 (Homotopy, CW complexes, and vector bundles). Define pointed homotopy, homotopy equivalence, loop and suspension spaces, homotopy groups, and CW complexes. Prove cellular approximation and the long exact homotopy sequence of a fibration. Define a topological complex vector bundle intrinsically by local product structure and fiberwise linear operations. Construct the infinite Grassmannian from its finite-dimensional subspace functor and prove that homotopy classes $[X, BU(n)]$ classify rank- n bundles over a finite CW complex. Define $K^0(X)$ by group completion under direct sum, prove homotopy invariance and the splitting principle, and derive its reduced suspension and clutching descriptions. When X is smooth, define a smooth vector bundle and prove by approximation and partitions of unity that every topological bundle is isomorphic to one with compatible smooth structure.

Exercise 22.2 ($U(1)$ gauge symmetry and characteristic classes). For the smooth vector bundles just constructed, define connections, curvature, and gauge transformations intrinsically. From invariant polynomials and the Bianchi identity prove that $\text{Tr exp}(iF/2\pi)$ is closed and connection-independent in cohomology. Construct integral Chern classes from transition functions and prove that their real images agree with Chern–Weil forms. Use the splitting principle to derive the Chern character and Todd and $\hat{A}$ classes and prove their multiplicative identities.

Exercise 22.3 (Fredholm operators and elliptic symbols). Define Fredholm operators and their analytic index, prove stability under compact perturbations, and construct the symbol class of an elliptic operator in $K_c^0(T^*M)$. Construct the bounded transform and prove that its index depends only on the compactly supported symbol class.

Exercise 22.4 (Bott periodicity and the topological index). Construct the harmonic-oscillator Clifford symbol on a complex vector space, compute its one-dimensional kernel, and prove that external product with its class gives the Bott isomorphism by radial contraction and Mayer–Vietoris. Construct Thom isomorphisms by the same Clifford class and define the topological index by embedding M in Euclidean space. Prove independence of the embedding by tubular neighborhoods and isotopy.

Exercise 22.5 (Atiyah–Singer index equality). Prove equality of analytic and topological indices first for the Bott operator, then for external products and open excision, and finally for every compact manifold by embedding and retraction. Apply the splitting principle and Chern–Weil theory to deduce

$$\text{ind } \bar{\partial}_E = \int_M \text{ch}(E) \text{Td}(T^{1,0}M)$$

on a compact complex manifold, and derive the $\hat{A}$ formula for a spin Dirac operator.

Exercise 22.6 (K3 elliptic genus). Define a K3 surface as a simply connected compact complex surface with trivial canonical bundle. Apply the Dolbeault index formula to the trivial bundle, use Serre duality, and prove $\int_{K3} c_2 = 24$ and $\hat{A}(K3) = 2$. With $y = e^{2\pi iz}$ and $q = e^{2\pi i\tau}$, define its elliptic genus by

$$y^{-1} \chi \left(K3, \bigotimes_{n \geq 1} \Lambda_{-yq^{n-1}} T^* \otimes \Lambda_{-y^{-1}q^n} T \otimes S_{q^n} T^* \otimes S_{q^n} T \right).$$

Here Λ_t and S_t are the total exterior- and symmetric-power operations in completed K -theory. Extend the theta constants to Jacobi theta functions by inserting $e^{2\pi i(n+a)z}$ in their characteristic lattice sums, and thereby define $\vartheta_1(z; \tau)$. Put

$$\phi_{0,1}(z, \tau) = 4 \left[\frac{\vartheta_2(z; \tau)^2}{\vartheta_2(0; \tau)^2} + \frac{\vartheta_3(z; \tau)^2}{\vartheta_3(0; \tau)^2} + \frac{\vartheta_4(z; \tau)^2}{\vartheta_4(0; \tau)^2} \right].$$

Use Riemann–Roch to derive the Chern–root theta product. Expand it, integrate, and prove $\text{Ell}_{K3}(z, \tau) = 2\phi_{0,1}(z, \tau)$. Derive its first Fourier–Jacobi coefficients from the theta identities already established.

Exercise 22.7 (Monopole harmonics). Let $L_N \rightarrow S^2$ be the homogeneous line bundle associated with the weight N character of $U(1) \subset SU(2)$. Use Peter–Weyl, rather than polar gauges, to decompose $L^2(S^2, L_N)$ into the $SU(2)$ representations V_j , $j = |N|/2, |N|/2 + 1, \dots$. Define the monopole harmonics as the corresponding distinguished matrix coefficients and derive their orthogonality, completeness, addition kernel, and magnetic-Laplacian values

$$\lambda_j = \frac{1}{R^2} \left(j(j+1) - \frac{N^2}{4} \right).$$

Identify the lowest level and its degeneracy $|N| + 1$ with Riemann–Roch.

Exercise 22.8 (Integer quantum Hall conductance). For a gapped family of Fermi projections $P(k)$ over the Brillouin torus, derive the Kubo response from adiabatic perturbation theory and prove

$$\sigma_{xy} = \frac{e^2}{h} \frac{i}{2\pi} \int_{\mathbb{T}^2} \text{Tr}(P \, dP \wedge dP) = C \frac{e^2}{h}, \quad C \in \mathbb{Z}.$$

Relate C to spectral flow by applying the index theorem to a flux insertion. For filled monopole-harmonic Landau levels, compute C from the associated line bundle and predict the Hall plateau and level degeneracy. Separate the gap, noninteracting-band, zero-temperature, and disorder-localization assumptions from the measured transverse conductance.

23 Cobordism

Exercise 23.1 (Spectra and stable homotopy). Using pointed CW complexes and the loop–suspension adjunction already developed, construct spectra and stable homotopy groups by iterated suspension, prove the Freudenthal stable range, and identify ordinary homology as maps into Eilenberg–Mac Lane spectra. Construct generalized homology and cohomology from a spectrum and prove their exactness and suspension axioms.

Exercise 23.2 (SO , Spin, and U structures). Define a tangential G -structure as a lift of the stable tangent classifier through $BG \rightarrow BO$. Construct orientation, spin, and stable complex structures as the cases SO , Spin, and U , and identify their obstructions w_1 , w_2 , and the stable complex lifting condition. Define G -bordism by compact manifolds with compatible boundary structure and prove it is a graded ring under disjoint union and product.

Exercise 23.3 (Pontryagin–Thom decomposition). For an embedding of a closed G -manifold in Euclidean space, collapse the complement of a tubular neighborhood and obtain a map to the Thom space of its normal bundle. Prove independence under stabilization and bordism. Conversely, make a stable map transverse to the zero section and recover a G -manifold. Prove the isomorphism between G -bordism and the stable homotopy groups of the Thom spectrum MG . Derive complex orientations and the formal group law of MU from tensor product of line bundles.

Exercise 23.4 (Eta functions and the APS boundary correction). For a self-adjoint elliptic operator D with discrete nonzero eigenvalues, define

$$\eta_D(s) = \sum_{\lambda \neq 0} \text{sign}(\lambda) |\lambda|^{-s}$$

for $\Re s$ large. Use the heat kernel Mellin transform to continue it to $s = 0$ and prove regularity there. Attach a cylinder to a compact manifold with boundary, analyze separated boundary modes, and prove

$$\text{ind } D_{\text{APS}}^+ = \int_M \hat{A}(TM) \text{ch}(E) - \frac{\eta_{D_\partial}(0) + \dim \ker D_\partial}{2}.$$

For the circle operator with holonomy $e^{2\pi i a}$ derive $\eta(0) = 1 - 2a$ for $0 < a < 1$ from the Hurwitz-zeta functional equation.

Exercise 23.5 (Higher categories and bordisms). Define an (∞, n) -category as a complete n -fold Segal space: state the Segal composition maps, completeness condition, and the requirement that k -morphisms above level n be invertible. Define symmetric monoidal structures by commutative algebra objects in this model and construct the (∞, n) -category whose objects are compact 0-manifolds and whose k -morphisms are collared k -bordisms, diffeomorphisms, isotopies, and higher isotopies. Prove associativity of gluing from contractibility of collar choices and identify disjoint union as its symmetric tensor product.

Exercise 23.6 (Fully extended field theories). Define adjoints for k -morphisms in a symmetric monoidal (∞, n) -category and call an object fully dualizable when it has evaluation and coevaluation at every level with all required adjoints. Construct framed bordisms by handles and prove that evaluation on a point gives an equivalence between fully extended framed topological field theories and fully dualizable objects. Add homotopy fixed-point data for a tangential G -structure. Verify in dimension one that the snake identities defining finite-dimensional vector spaces at the beginning of the manuscript are exactly the required point data.

Exercise 23.7 (Invertible phases and eta-response predictions). Define an invertible reflection-positive field theory as a symmetric monoidal map from the appropriate bordism category to invertible complex lines, and use stacking to form its classification group. Apply Pontryagin–Thom to identify deformation classes with maps from the relevant Thom spectrum to the Picard spectrum of invertible phases, including the symmetry structure in the tangential data. For a massive fermion whose boundary operator is D_∂ , derive the relative partition phase

$$Z_{\text{top}} = \exp \left[-\frac{\pi i}{2} (\eta_{D_\partial}(0) + \dim \ker D_\partial) \right].$$

Use the circle calculation to predict its change under one unit of flux and the resulting protected boundary spectral flow. State locality, gap, symmetry, and interaction assumptions separately from the measurable phase and crossing parity.

Part IX

Tensor Categories

24 Geometric Satake

Exercise 24.1 (Loop groups and Schubert cells). For a complex reductive group G , define $G((z))$, $G[[z]]$, the faithfully flat quasi-compact topology on affine schemes, and sheafification of a functor of points. Define the fpqc quotient $\text{Gr}_G = G((z))/G[[z]]$. Prove the Cartan decomposition

$$\text{Gr}_G = \bigsqcup_{\lambda \in X_*(T)_+} G[[z]] z^\lambda$$

and compute the orbit dimension $\langle 2\rho, \lambda \rangle$. Construct the Schubert varieties by orbit closure and derive their dominance order and convolution correspondence from modification of G -bundles on a formal disk.

Exercise 24.2 (Constructible derived categories and Verdier duality). For a finite stratification of a locally compact space, define bounded constructible complexes of sheaves and chain homotopies. Localize the homotopy category at quasi-isomorphisms to construct the constructible derived category. Resolve by injective and soft sheaves to construct Rf_* , $Rf_!$, Lf^* , $f^!$, $\otimes^L$, and $R\mathcal{H}om$; derive their adjunctions, the projection formula, and proper base change from the corresponding statements for the resolutions. Construct the dualizing complex from compactly supported cochains and define Verdier duality $\mathbb{D}\mathcal{F} = R\mathcal{H}om(\mathcal{F}, \omega_X)$. Prove biduality and the interchange of Rf_* with $Rf_!$ under $\mathbb{D}$.

Exercise 24.3 (Perverse sheaves and convolution). For the Schubert stratification, define the perverse truncation by the support and cosupport inequalities. Prove that intersection cohomology extensions are simple perverse sheaves. Define convolution through the proper multiplication correspondence, prove perversity by semismallness, and derive associativity, commutativity from fusion on two moving points, and rigid duals from inversion and Verdier duality.

Exercise 24.4 (Satake fiber functor). Let Sat_G be the category of $G[[z]]$ -equivariant perverse sheaves on Gr_G . Prove exactness and faithfulness of total cohomology by the weight-functor decomposition over semi-infinite orbits. Use convolution to make it a symmetric monoidal fiber functor. Construct

the group of its tensor automorphisms, recover its coordinate Hopf algebra from matrix coefficients and rigid duals, and prove the Tannaka equivalence with representations of a reductive group.

Exercise 24.5 (Geometric Satake equivalence). Compute the roots of this Tannaka group from the rank-one Schubert varieties and prove that its root datum is dual to that of G . Deduce the monoidal equivalence

$$\text{Sat}_G \simeq \text{Rep}(G^\vee)$$

and identify intersection cohomology of $\overline{\text{Gr}}_\lambda$ with the irreducible highest-weight representation V_λ .

Exercise 24.6 (Hall–Littlewood polynomials). For a partition λ define

$$P_\lambda(x; t) = \frac{1}{v_\lambda(t)} \sum_{w \in S_r} w \left(x^\lambda \prod_{i < j} \frac{x_i - tx_j}{x_i - x_j} \right)$$

with $v_\lambda(t)$ fixed by monicity. Prove polynomiality, triangularity, orthogonality, the Cauchy identity, and the Pieri rule. Over a finite field, stratify the convolution fibers in Gr_{GL_r} and show that their point counts give the Hall–Littlewood structure constants at $t = |\mathbb{F}|^{-1}$. At $t = 0$ recover the Weyl characters already developed for reductive groups.

Exercise 24.7 ('t Hooft-line fusion). Label magnetic line operators in a four-dimensional gauge theory with gauge group G by dominant coweights. Regard them as codimension-three defects in the extended field-theory framework of Cobordism. Model their collision by convolution of Schubert cycles and use geometric Satake to prove

$$L_\lambda L_\mu = \sum_\nu \dim \text{Hom}_{G^\vee}(V_\nu, V_\lambda \otimes V_\mu) L_\nu.$$

Compute these integers from Hall–Littlewood multiplication and specialize to $G = PGL_2$ to obtain the full $SU(2)$ triangle rule. Thus predict the allowed magnetic charges and their multiplicities after two line defects fuse; separate topological protection and operator normalization from the measured fusion channels.

25 Fusion Categories

Exercise 25.1 (Rigid and unitary fusion categories). Define a unitary fusion category as a finite semisimple rigid $\mathbb{C}$ -linear C^* -tensor category with simple tensor unit. Define evaluation, coevaluation, pivotal and spherical structures, quantum dimensions, and the global dimension. Derive the snake, rigidity, Frobenius reciprocity, and $\sum_c d_{ab}^c d_c = d_a d_b$ identities intrinsically. Define braidings, twists, and nondegeneracy through the categorical trace of double braiding. Show that the line-defect fusion category from the preceding section is the prototype, while noninvertible simple objects require no underlying group.

Exercise 25.2 (Temperley–Lieb categories and fusion semisimplification). Define TL_d by planar pairings and loop value d , and prove the tensor and $*$ structures by gluing diagrams. Construct the Jones–Wenzl idempotents recursively and derive $f_1 \otimes f_n \simeq f_{n-1} \oplus f_{n+1}$. At $d = 2 \cos(\pi/(k+2))$, identify the negligible ideal generated by f_{k+1} and prove that the semisimplification has simples $0, \dots, k$ and fusion

$$a \otimes b = \bigoplus_{\substack{c=|a-b| \\ c \leq \min(a+b, 2k-a-b)}}^{(2)} c.$$

Construct the ribbon functor to the root-of-unity tilting quotient developed in Quantum Groups by sending one strand to V_1 , prove that its kernel is the negligible ideal generated by f_{k+1} , and identify the two semisimplifications.

Exercise 25.3 (Factorization homology and the Levin–Wen model). Define a disk algebra in a symmetric monoidal higher category and construct factorization homology by the colimit over embedded disks. Prove excision by cofinality of collar gluings. Apply it to the three-dimensional Turaev–Viro theory of a fusion category $\mathcal{C}$ and identify the category assigned to a circle with its Drinfeld center. Triangulate a closed surface, define the Levin–Wen vertex and plaquette projectors from the F -symbols, and prove their commutativity by the pentagon identity. Show that the ground-state space is the corresponding factorization-homology state space. Verify explicitly that $\mathcal{C} = SU(2)_1$ is the semion category and its Levin–Wen theory is doubled semion, not the toric code.

Exercise 25.4 (Modular S -matrix and the Verlinde decomposition). For $\mathcal{C} = SU(2)_k$, compute the categorical trace of double braiding from the root-of-unity ribbon structure constructed in Quantum Groups and recover the finite sine transform obtained from affine characters:

$$S_{ab} = \sqrt{\frac{2}{k+2}} \sin\left(\frac{(a+1)(b+1)\pi}{k+2}\right), \quad \theta_a = e^{\pi i a(a+2)/(2(k+2))}.$$

Prove directly that the two constructions agree, then use its orthogonality to deduce nondegeneracy of braiding and the Verlinde formula

$$N_{ab}^c = \sum_x \frac{S_{ax} S_{bx} \overline{S_{cx}}}{S_{0x}}.$$

Recover the truncated fusion rule and compute the torus ground-state degeneracy. Relate these trigonometric special functions to the root-of-unity limit of the quantum-group characters without redefining them.

Exercise 25.5 (Anyon interferometry). For simple anyons a, b , prove that a probe a encircling total charge b has normalized monodromy

$$M_{ab} = \frac{S_{ab} S_{00}}{S_{0a} S_{0b}}.$$

In a balanced two-path interferometer with controllable phase ϕ , derive

$$P_{\pm}(b) = \frac{1}{2} \left(1 \pm \Re(e^{i\phi} M_{ab}) \right).$$

Insert the $SU(2)_k$ sine matrix to predict every fringe displacement and visibility and show which charges are distinguishable by a fixed probe. State coherence, braiding adiabaticity, charge-superselection, path balance, and detector-resolution assumptions separately from these probabilities.

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